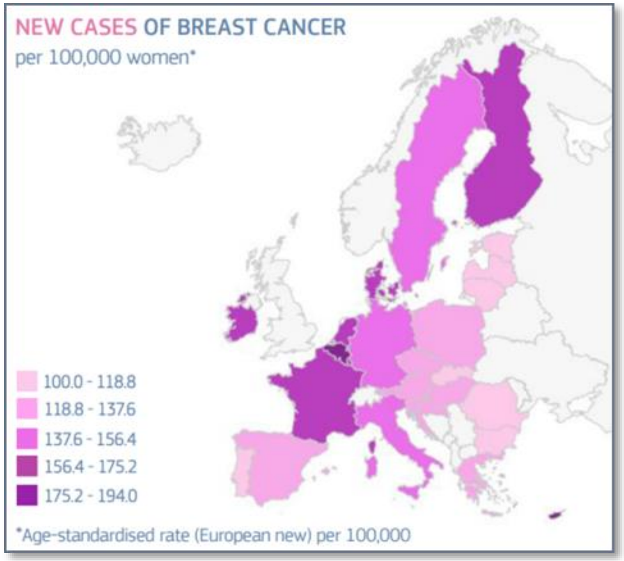
Evaluation of the technical feasibility of producing Breast Cancer Polygenic Risk Scores in a clinical genetics routine laboratory

Marie BRUNEAU
Genetic Counsellor
ULB Center of Human Genetics

BREAST CANCER

- ☐ Breast cancer is the most frequently occurring cancer with 355,457 new cases per year in Europe, it represent 13,3% of all new cancer cases.

- ☐ Prevalence:
  - 14% in Europe
  - 12% in the world



https://ecis.jrc.ec.europa.eu/pdf/Breast_cancer_factsheet-Oct_2020.pdf

Complex and heterogeneous disease

Genetic aspects

Lifestyle

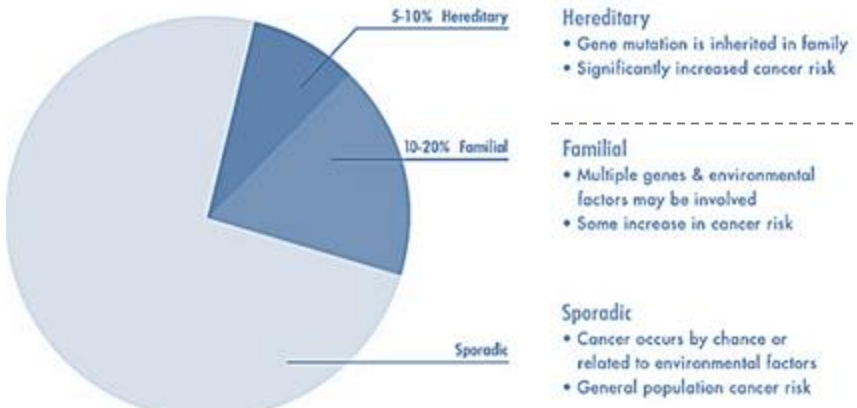
Environmental exposures

Systemic health features

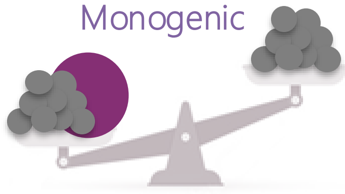
GENETIC CANCER

Breast cancer risk assessment is a critical part of public health.

The identification of high-risk patients is a major challenge in order to adapt follow-up and increase the chances of early detection of the disease.



Monogenic



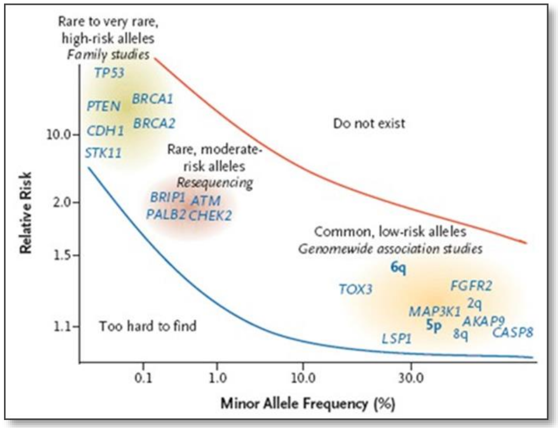
1 gene = 1 effect

Genetic tests proposed in clinical routine (according to inclusion criteria) to identify patients at risk. Genes frequently involved in monogenic forms: *BRCA1/2*, *PALB2*, *CHEK2* and *ATM*

Polygenic

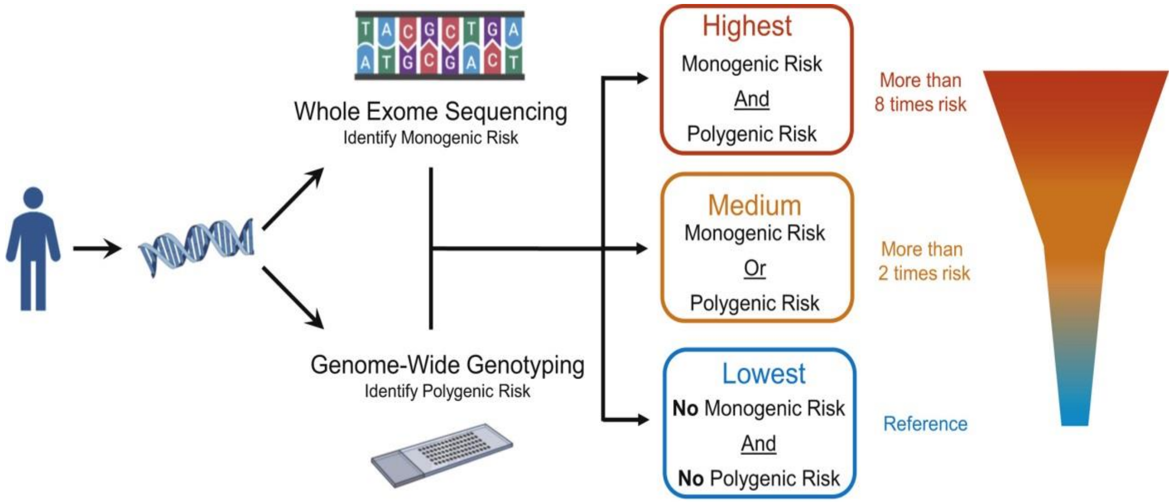
Several genes = 1 effect

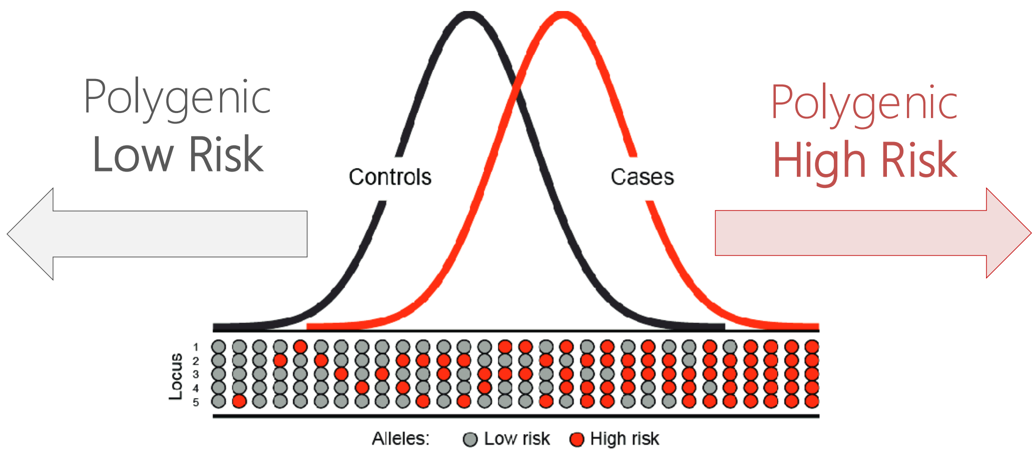
A large proportion of familial cancer cases are not explained by mendelian genetics and are attributable to polygenic risk factors. These tests are currently not available in clinical routine.



POLYGENIC RISK SCORE

A polygenic risk score (PRS) estimates the genetic risk of an individual for some disease or trait, calculated by aggregating the effect of many common variants associated with the condition.





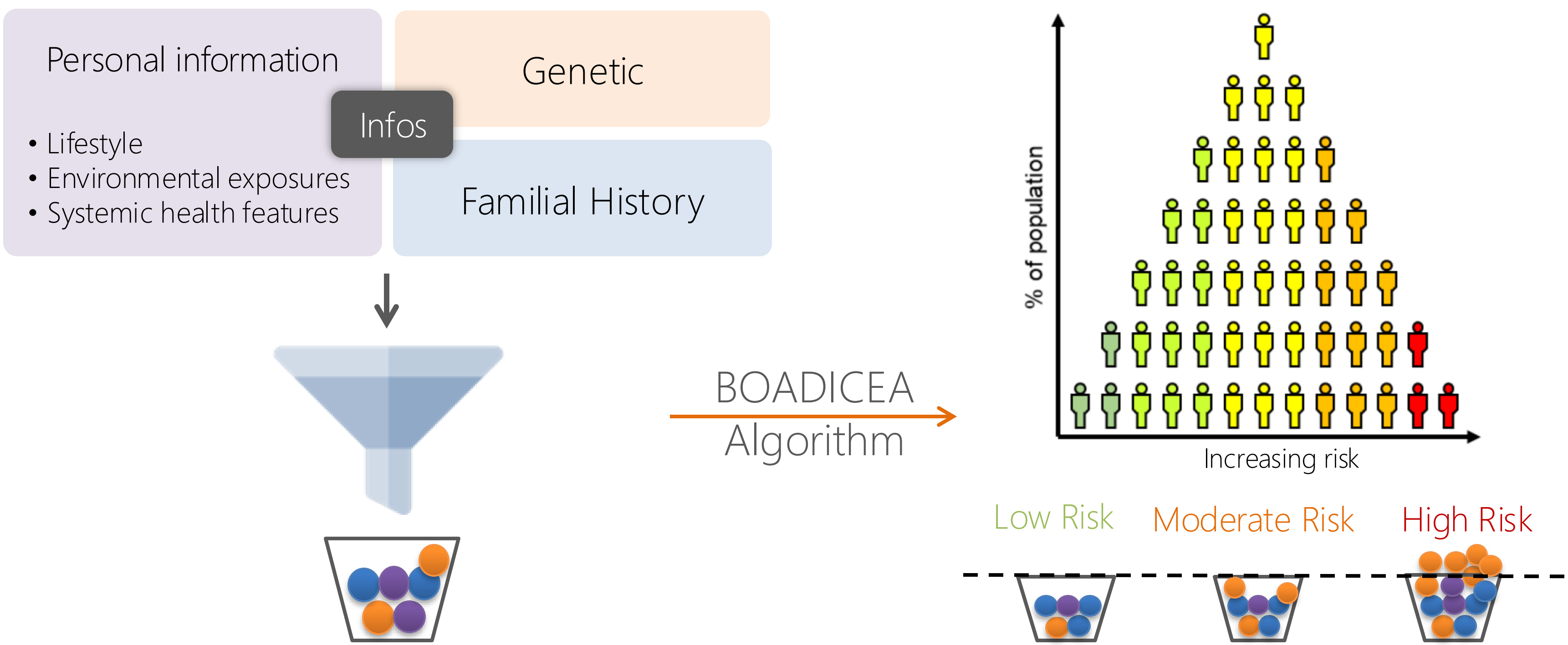
Canrisk - BOADICEA

BOADICEA (Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm) is a risk calculator for breast cancer.

It systematically identifies those at highest and lowest risk.

It has been developed by University of Cambridge, UK and is free of access for everyone.

It is continuously validated by multiple scientific publications.



The algorithm allows for the integration of multiple genetic and non-genetic risk factors.

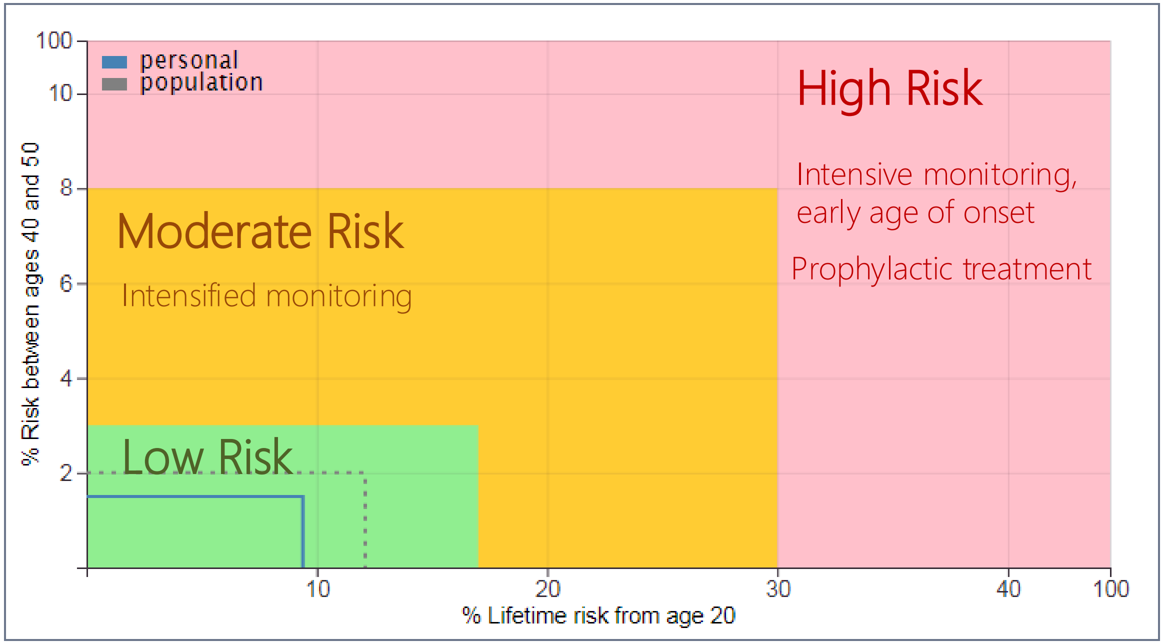
It systematically identifies those at highest and lowest risk.

Canrisk - BOADICEA

In order to be more accessible to the public and usable in clinics. The BOADICEA program is used via the Canrisk web interface.

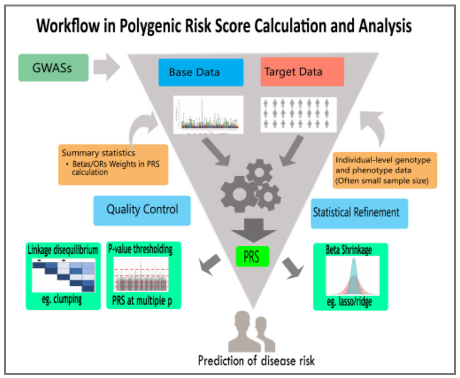
The result is transmitted in the form of a table or a graph and gives a risk stratification of the patient

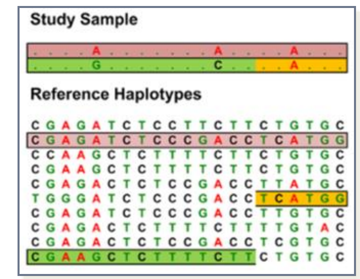
Patient risk stratification, taking into account genetic and non-genetic risk factors



POLYGENIC RISK SCORE

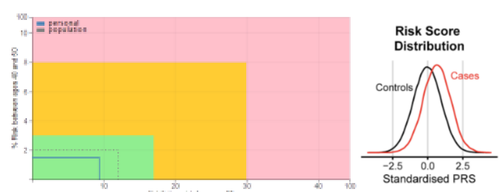
VALIDATED POLYGENIC RISK SCORE → SNP ARRAY → QC AND FILE FORMATTING → IMPUTATION





QC AND FILE FORMATTING → CLINICAL ALGORITHM → INTERPRETATION → GENETIC COUNSELLING

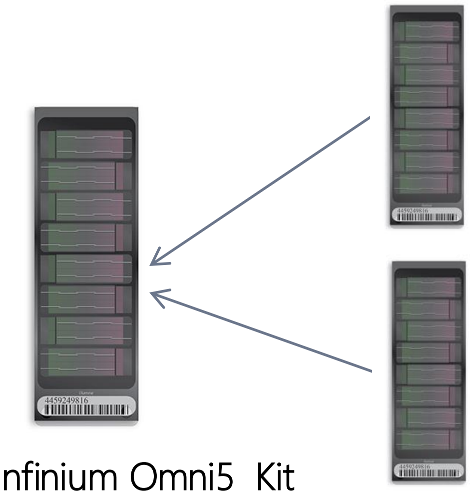
CanRisk



Available technology

iScan System - illumina

Infinium Omni2.5 Kit



Infinium Omni2.5

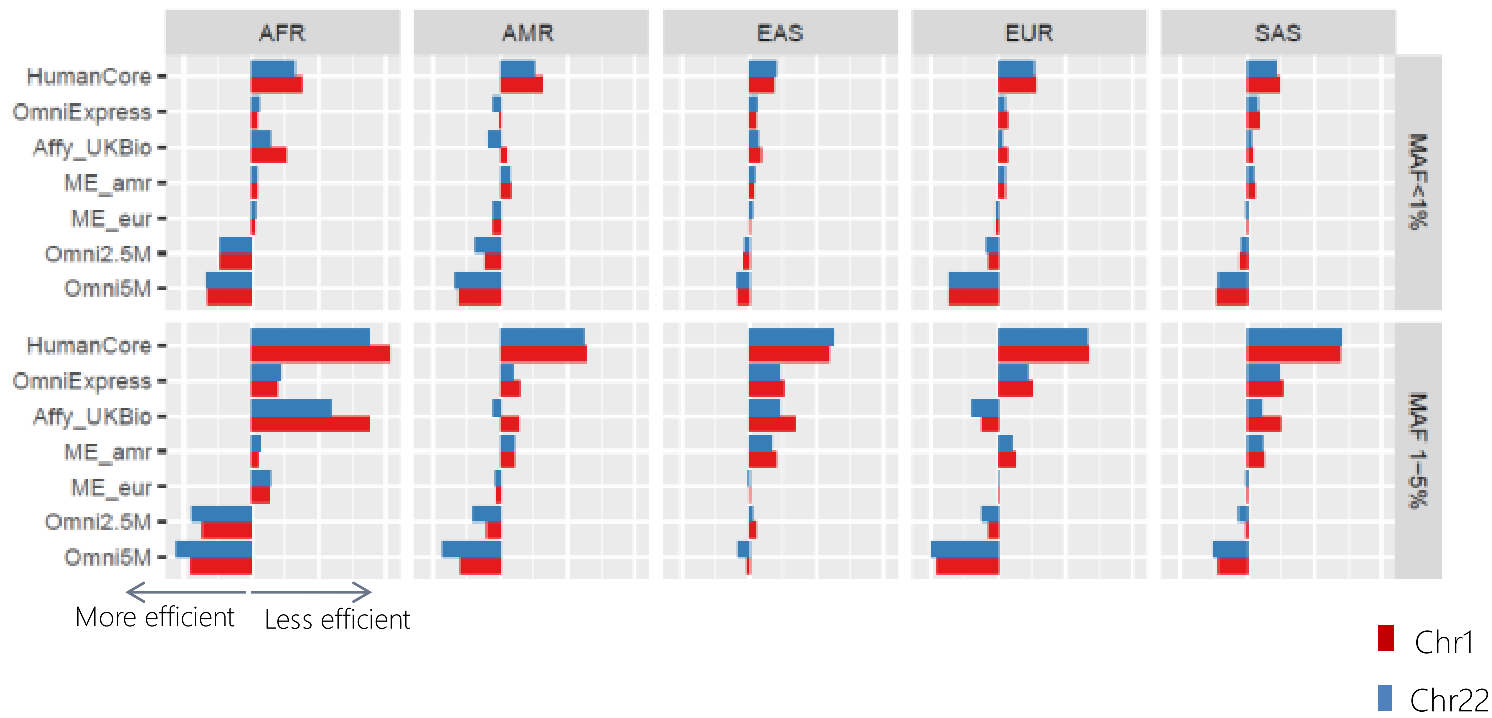
Infinium Omni2.5S

Infinium Omni5 Kit

	Infinium Omni2.5 Kit	Infinium Omni5 Kit
	☐ Comprehensive set of both common and rare single nucleotide polymorphism (SNP)	☐ High-density array coverage of common, intermediate, and rare SNPs while leveraging powerful tag SNPs
Number of Markers	☐ 2,381,000	☐ 4,284,426
Minor Allele Freq. Range	☐ >2,5% MAF	☐ >1% MAF
Content Source	☐ tagSNPs selected from 1000 Genomes Project	☐ tagSNPs selected from the International HapMap and 1000 Genomes Projects

POLYGENIC RISK SCORE - Microarrays

COMPARISON OF SNP ARRAYS TO DISCRIMINATE DIFFERENT POPULATIONS



Omni 5 and 2.5 seem to be the best arrays, with the Omni 5 marginally better than 2.5.

POLYGENIC RISK SCORE - Microarrays

OMNI 2.5	Patients
205992600009_R01C01	Euro 1
205992600009_R02C01	AfroSub1
205992600009_R03C01	AfroSub2
205992600009_R04C01	Hisp1
205992600009_R05C01	Euro2
205992600009_R06C01	Euro3
205992600009_R07C01	Euro4
205992600009_R08C01	Euro5

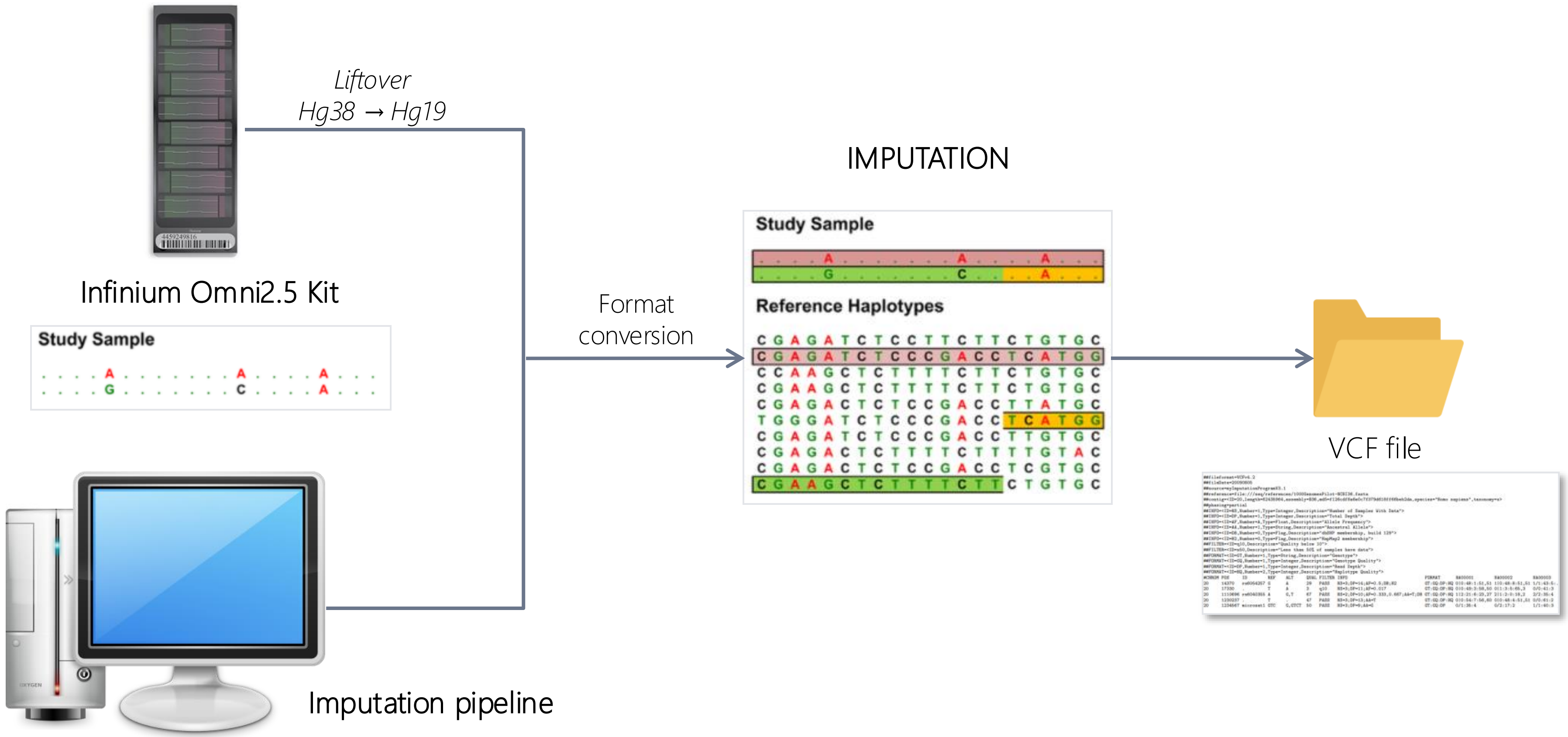
Patients - Test

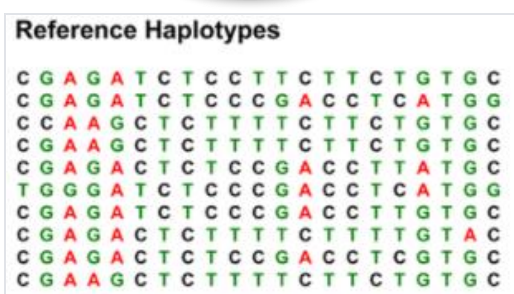
- 8 patients were tested, both with the Omni 2.5 and 5 arrays
- Ethnic origin Europe, Hispanic and Sub-Saharan Africa
- Medical history of oncology not known (random samples)
- Only OMNI 2.5 data have been studied at the present time

Infinium Omni2.5 Kit

POLYGENIC RISK SCORE - Imputation

Genotype imputation is a process of estimating missing genotypes from the haplotype or genotype reference panel. It can effectively boost the power of detecting single nucleotide polymorphisms (SNPs) in genome-wide association studies, integrate multi-studies, produce fine-mapping studies and is necessary to deduce the useful PRS SNPs in a clinical setting.





Trans-Omics for Precision Medicine (TopMed) imputation webserver

- Algorithm minimac4 used for imputation
- Reference panel including information from 97,256 deeply sequenced human genomes.
- >30X coverage

POLYGENIC RISK SCORE - Imputation

The VCF file contains the data of the 8 patients and the data concerning the quality of the imputation.

Meta-information lines

Contain information of :
- Format of File
- Filter
- Assembly
- Pedigree
- Sample

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##FORMAT=<ID=DS,Number=1,Type=Float,Description="Estimated Alternate Allele Dosage : [P(0/1)+2*P(1/1)]">
##FORMAT=<ID=HDS,Number=2,Type=Float,Description="Estimated Haploid Alternate Allele Dosage">
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##originalFile=infinium-omni-25-imputed-topmed.vcf.gz
##targetRefGenome=/home/cdetry/references/human_g1k_v37.fasta
##liftOverDate=May13,2022
```

Header

#CHROM : Chromosome number
POS: Position of the SNP
ID: ID of SNP
REF : Reference allele
ALT : Alternative allele
QUAL : Quality
FILTER
INFO :
- AF: effect allele frequency
- MAF : minor allele frequency
- R2 : indicates the amount of phenotypic variance explained by the PRS
- IMPUTED / TYPED :

FORMAT

<table>
<tr><th>#CHROM</th><th>POS</th><th>ID</th><th>REF</th><th>ALT</th><th>QUAL</th><th>FILTER</th><th>INFO</th><th>FORMAT</th><th>205992600009_R01C01</th><th>205992600009_R02C01</th><th>20599260</th><th>20599260</th><th>20599260</th><th>20599260</th><th>20599260</th><th>205992600009_R08C01</th></tr>
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<tr><td>1</td><td>114445880</td><td>chr1:113903258:G:A</td><td>G</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.1249</td><td>GT</td><td>0|0</td><td>0|0</td><td>0|0</td><td>0|0</td><td>1|0</td><td>0|0</td><td>1|0</td><td>0|0</td></tr>
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<tr><td>2</td><td>19315675</td><td>chr2:19115914:T:A</td><td>T</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.3747</td><td>GT</td><td>1|0</td><td>0|1</td><td>0|1</td><td>0|0</td><td>0|0</td><td>0|1</td><td>1|0</td><td>0|1</td></tr>
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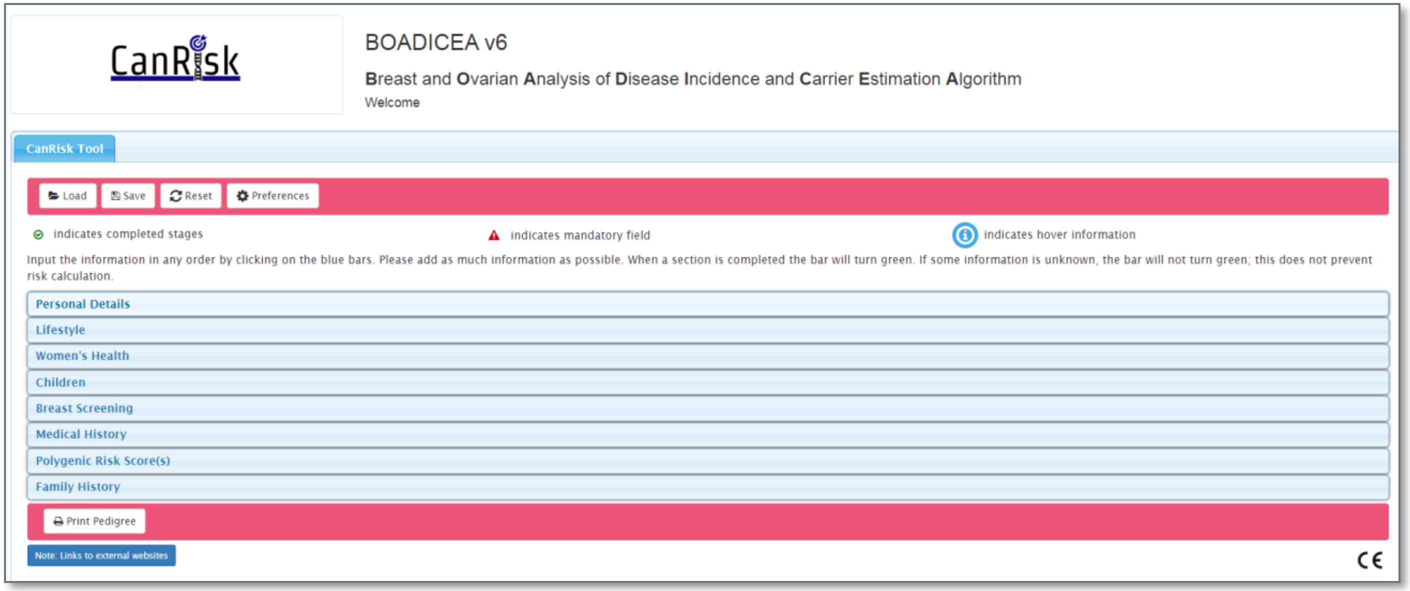
</table>

CanRisk Tool

CanRisk Interface

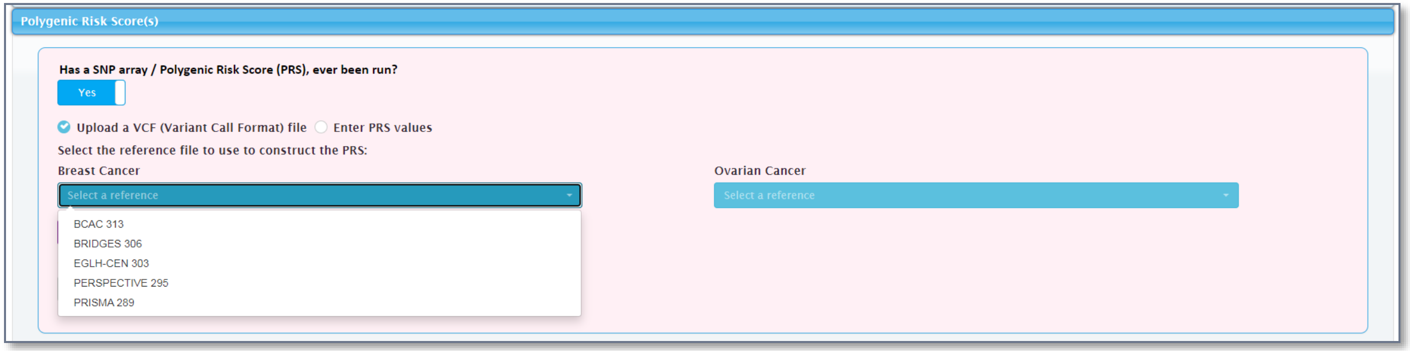
Allows for the integration of
- personal information
- Familial history
- Genetic (panel, PRS)

Each element provided will be integrated into the risk calculation



The Canrisk tool allows the use of several PRS

We used the BCAC313. It is one of the most used nowadays and has benefited from numerous validation studies.



CanRisk Tool – PRS

PRS BCAC313

- ☐ 313 SNPs :
  - 152 with increasing risk effect 48%
  - 161 with protecting effect 51%
- ☐ Development dataset comprised 94,075 case subjects and 75,017 control subjects of European ancestry
- ☐ 69 studies, divided into training and validation sets

Polygenic Risk Scores for Prediction of Breast Cancer and Breast Cancer Subtypes

Nasim Mavaddat,[1,*] Kyriaki Michailidou,[1,2] Joe Dennis,[1] Michael Lush,[1] Laura Fachal,[3] Andrew Lee,[1] Jonathan P. Tyrer,[3] Ting-Huei Chen,[4] Qin Wang,[1] Manjeet K. Bolla,[1] Xin Yang,[1] Muriel A. Adank,[5] Thomas Ahearn,[6] Kristiina Aittomäki,[7] Jamie Allen,[1] Irene L. Andrulis,[8,9] Hoda Anton-Culver,[10] Natalia N. Antonenkova,[11] Volker Arndt,[12] Kristan J. Aronson,[13] Paul L. Auer,[14,15] Päivi Auvinen,[16,17,18] Myrto Barrdahl,[19] Laura E. Beane Freeman,[6] Matthias W. Beckmann,[20] Sabine Behrens,[19] Javier Benitez,[21,22] Marina Bermisheva,[23] Leslie Bernstein,[24] Carl Blomqvist,[25,26] Natalia V. Bogdanova,[11,27,28] Stig E. Bojesen,[29,30,31] Bernardo Bonanni,[32] Anne-Lise Børresen-Dale,[33,34] Hiltrud Brauch,[35,36,37] Michael Bremer,[27] Hermann Brenner,[12,37,38] Adam Brentnall,[39] Ian W. Brock,[40] Angela Brooks-Wilson,[41,42] Sara Y. Brucker,[43] Thomas Brüning,[44] Barbara Burwinkel,[45,46] Daniele Campa,[19,47] Brian D. Carter,[48] Jose E. Castelao,[49] Stephen J. Chanock,[6] Rowan Chlebowski,[50] Hans Christiansen,[27] Christine L. Clarke,[51] J. Margriet Collée,[52] Emilie Cordina-Duverger,[53] Sten Cornelissen,[54] Fergus J. Couch,[55] Angela Cox,[40] Simon S. Cross,[56] Kamila Czene,[57]

(Author list continued on next page)

Stratification of women according to their risk of breast cancer based on polygenic risk scores (PRSs) could improve screening and prevention strategies. Our aim was to develop PRSs, optimized for prediction of estrogen receptor (ER)-specific disease, from the largest available genome-wide association dataset and to empirically validate the PRSs in prospective studies. The development dataset comprised 94,075 case subjects and 75,017 control subjects of European ancestry from 69 studies, divided into training and validation sets. Samples were genotyped using genome-wide arrays, and single-nucleotide polymorphisms (SNPs) were selected by stepwise regression or lasso penalized regression. The best performing PRSs were validated in an independent test set comprising 11,428 case subjects and 18,323 control subjects from 10 prospective studies and 190,040 women from UK Biobank (3,215 incident breast cancers). For the best PRSs (313 SNPs), the odds ratio for overall disease per 1 standard deviation in ten prospective studies was 1.61 (95%CI: 1.57–1.65) with area under receiver-operator curve (AUC) = 0.630 (95%CI: 0.628–0.651). The lifetime risk of overall breast cancer in the top centile of the PRSs was 32.6%. Compared with women in the middle quintile, those in the highest 1% of risk had 4.37- and 2.78-fold risks, and those in the lowest 1% of risk had 0.16- and 0.27-fold risks, of developing ER-positive and ER-negative disease, respectively. Goodness-of-fit tests indicated that this PRS was well calibrated and predicts disease risk accurately in the tails of the distribution. This PRS is a powerful and reliable predictor of breast cancer risk that may improve breast cancer prevention programs.

Extract from the file containing the list of SNPs and the statistical information contributing to the calculation of the PRS

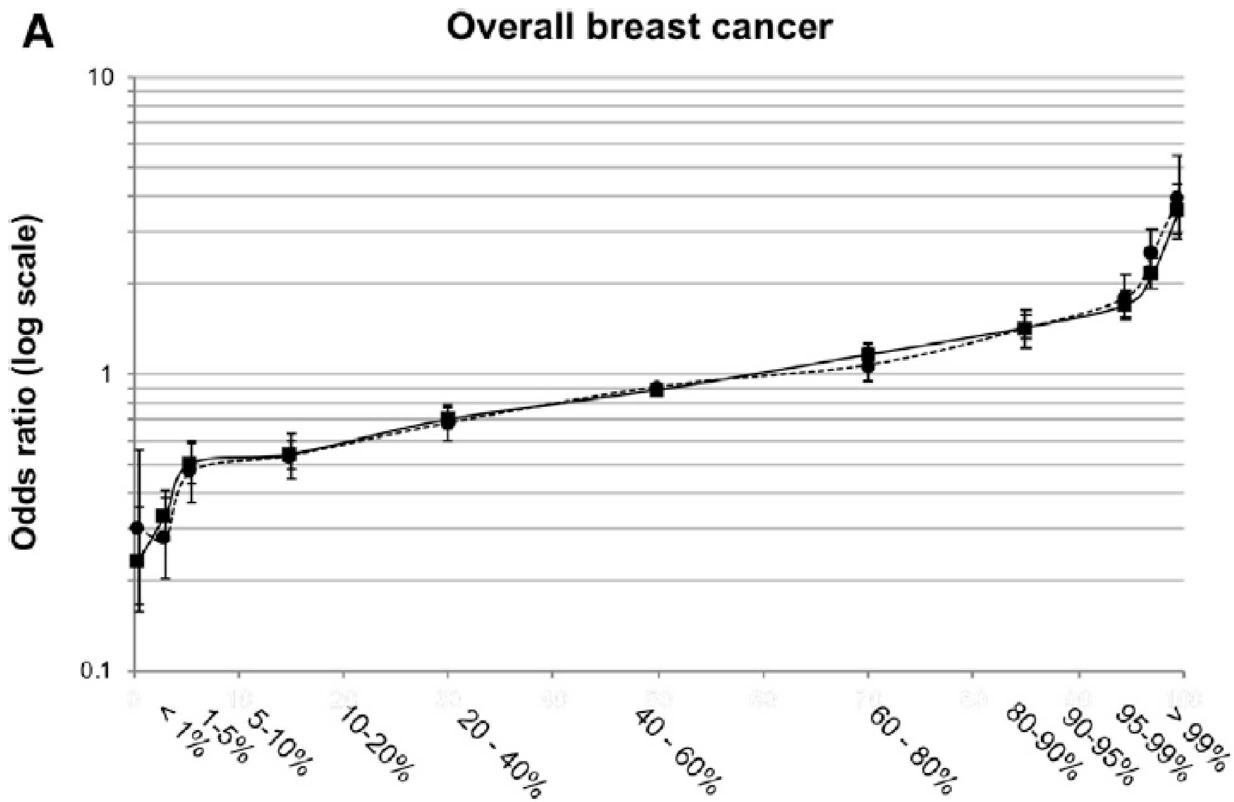
SNP[a]	Chromosome	Position[b]	Reference Allele	Effect Allele	EAF[c]	Overall Breast Cancer[d]	ER-positive[e]	ER-negative[f]	hybrid ER-positive[g]	hybrid ER-negative[h]
1_100880328_A_T	1	100880328	A	T	0,4097	0,0373	0,0355	0,016	0,0373	0,0373
1_10566215_A_G	1	10566215	A	G	0,3290	-0,0586	-0,0407	-0,1109	-0,0407	-0,1109
1_110198129_CAAA_C	1	110198129	CAAA	C	0,7755	0,0458	0,0545	0,0266	0,0458	0,0458
1_114445880_G_A	1	114445880	G	A	0,1664	0,0621	0,0642	0,0579	0,0621	0,0621
1_118141492_A_C	1	118141492	A	C	0,2657	0,0452	0,0417	0,0551	0,0452	0,0452

EAF : affect allele freq.

Lasso statistic coefficient values .

CanRisk Tool – PRS

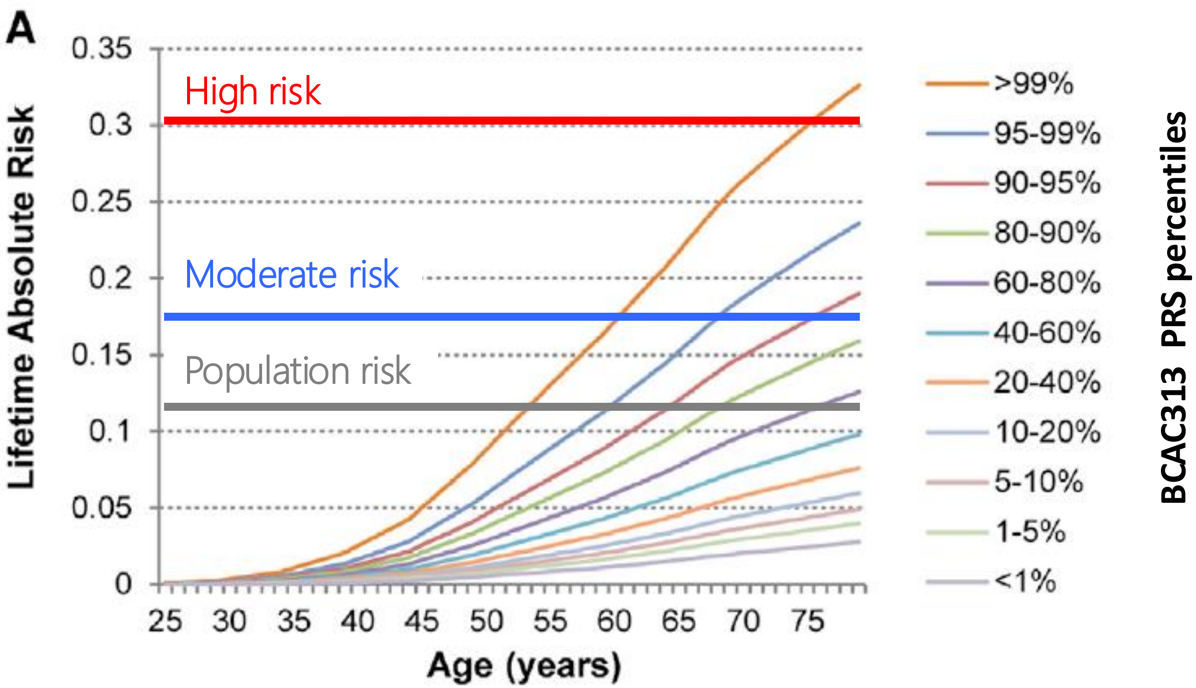
Patients with P>97 (+2 SD) or P< 3% (-2SD) have a substantial amplification or diminution of risk



BCAC313 PRS percentiles

CanRisk Tool – PRS

Cumulative lifetime absolute risk of developing breast cancer



Population, moderate and high risk thresholds are the ones described in the NICE UK guidelines and in slide 6.

CanRisk Tool – file formating issues

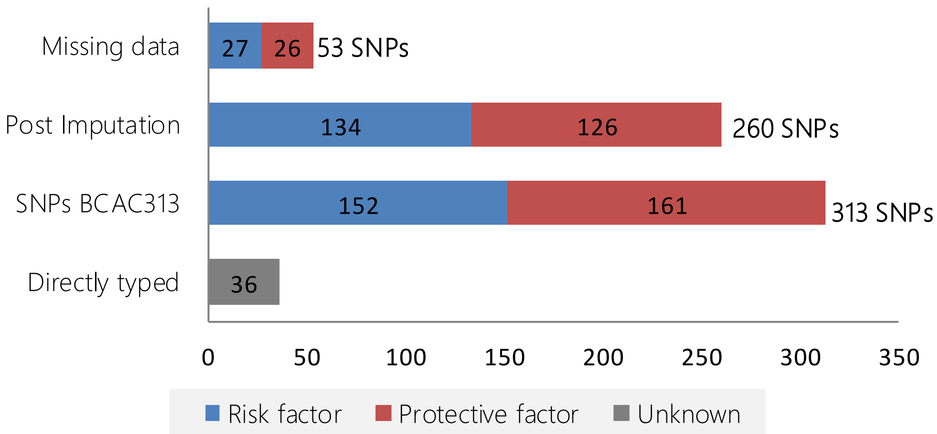
Missing Genotype

Error: The variant at position 84370124 on chromosome 4 was not found in the genotype file.
VCF File: corrige15_omni25-topmed-GTonly-PRSref.txt

Directly genotyped data, or imputed data may contain missing data. In our case we found that 53 SNPs out of the 313 of BCAC313 were missing (36 directly typed and 224 imputed).

Possible explanations:

- Loss of data during liftover



QUICK FIXES FOR THE PRESENT WORK:

- We have virtually added the missing SNPs with 0|0 genotype (homozygous wild type).
- The information concerning the MAF and AF are not filled in
- In this way we ensure that these SNPs do not contribute to the score
- Note that the missing SNPs are balanced between risk and protective effects.

Visualization of the data encoding

	A	B	C	D	E	F	G	H	I	J	K	L
103	#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	2059926000(	2059926000(	2059926000(
365	1	51467096	1:51467096:(	CT	.	.	.	AF=.;MAF=.;R2=.	GT	0|0	0|0	0|0
366	1	110198129	1:110198129	CAAA	.	.	.	AF=.;MAF=.;R2=.	GT	0|0	0|0	0|0
367	1	145604302	1:145604302	C	.	.	.	AF=.;MAF=.;R2=.	GT	0|0	0|0	0|0
368	1	168171052	1:168171052	CA	.	.	.	AF=.;MAF=.;R2=.	GT	0|0	0|0	0|0
369	1	172328767	1:172328767	T	.	.	.	AF=.;MAF=.;R2=.	GT	0|0	0|0	0|0

statistical information not provided

Genotype wild type

CanRisk Tool – file formating issues

Information reading

Error: In the method Vcf2Prs.validate_record for the record on chromosome chr1 at position 41389220, for sample 205992600009_R01C01 the dosage 0.035, and genotype 0|0 are inconsistent.
VCF File: infinium-omni-25-imputed-topmed-liftedbackhg19-313snps.vcf

<table>
<tr><th>#CHROM</th><th>POS</th><th>ID</th><th>REF</th><th>ALT</th><th>QUAL</th><th>FILTER</th><th>INFO</th><th>FORMAT</th><th>205992600009_R01C01</th><th>205992600009_R02C01</th></tr>
<tr><td>chr1</td><td>7917076</td><td>chr1:7857016:G:A</td><td>G</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.31172;MAF=0.31172;R2=0.99638;IMPUTED</td><td>GT:DS:HDS:GP</td><td>0|0:0:0,0:1,0,0</td><td>1|0:1.000:1.000,0.000:0.000,1.000,0.000</td></tr>
<tr><td>chr1</td><td>10566215</td><td>chr1:10506158:A:G</td><td>A</td><td>G</td><td>.</td><td>PASS</td><td>AF=0.38117;MAF=0.38117;R2=0.97642;ER2=0.79539;TYPED</td><td>GT:DS:HDS:GP</td><td>1|0:1.000:1.000,0.000:0.000,1.000,0.000</td><td>1|1:2.000:1.000,1.000:0.000,0.000,1.000</td></tr>
<tr><td>chr1</td><td>18807339</td><td>chr1:18480845:T:C</td><td>T</td><td>C</td><td>.</td><td>PASS</td><td>AF=0.56034;MAF=0.43966;R2=0.99141;IMPUTED</td><td>GT:DS:HDS:GP</td><td>0|0:0:0,0:1,0,0</td><td>1|0:0.974:0.974,0.000:0.026,0.974,0.000</td></tr>
<tr><td>chr1</td><td>41380440</td><td>chr1:40914768:C:T</td><td>C</td><td>T</td><td>.</td><td>PASS</td><td>AF=0.68750;MAF=0.31250;R2=1.00000;ER2=1.00000;TYPED</td><td>GT:DS:HDS:GP</td><td>1|1:2.000:1.000,1.000:0.000,0.000,1.000</td><td>0|0:0:0,0:1,0,0</td></tr>
<tr><td>chr1</td><td>41389220</td><td>chr1:40923548:T:C</td><td>T</td><td>C</td><td>.</td><td>PASS</td><td>AF=0.00222;MAF=0.00222;R2=0.03155;IMPUTED</td><td>GT:DS:HDS:GP</td><td>0|0:0.035:0.000,0.035:0.965,0.035,0.000</td><td>0|0:0:0,0:1,0,0</td></tr>
<tr><td>chr1</td><td>46670206</td><td>chr1:46204534:TC:T</td><td>TC</td><td>T</td><td>.</td><td>PASS</td><td>AF=0.24988;MAF=0.24988;R2=0.99934;IMPUTED</td><td>GT:DS:HDS:GP</td><td>0|0:0:0,0:1,0,0</td><td>1|0:0.999:0.999,0.000:0.001,0.999,0.000</td></tr>
<tr><td>chr1</td><td>88156923</td><td>chr1:87691240:G:A</td><td>G</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.06055;MAF=0.06055;R2=0.96680;IMPUTED</td><td>GT:DS:HDS:GP</td><td>0|1:0.969:0.000,0.969:0.031,0.969,0.000</td><td>0|0:0:0,0:1,0,0</td></tr>
<tr><td>chr1</td><td>88428199</td><td>chr1:87962516:C:A</td><td>C</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.25000;MAF=0.25000;R2=1.00000;ER2=1.00000;TYPED</td><td>GT:DS:HDS:GP</td><td>0|1:1.000:0.000,1.000:0.000,1.000,0.000</td><td>0|1:1.000:0.000,1.000:0.000,1.000,0.000</td></tr>
</table>

INFO :
- AF: effect allele frequency
- MAF : minor allele frequency
- R2 : indicates the amount of phenotypic variance explained by the PRS
- IMPUTED / TYPED

FORMAT:
- GT : Estimated most likely genotype
- GS : Estimated alternate allele dosage
- HDS :Estimated phased haploid alternate allele dosage
- GP : Estimated Posterior Genotype Probabilities

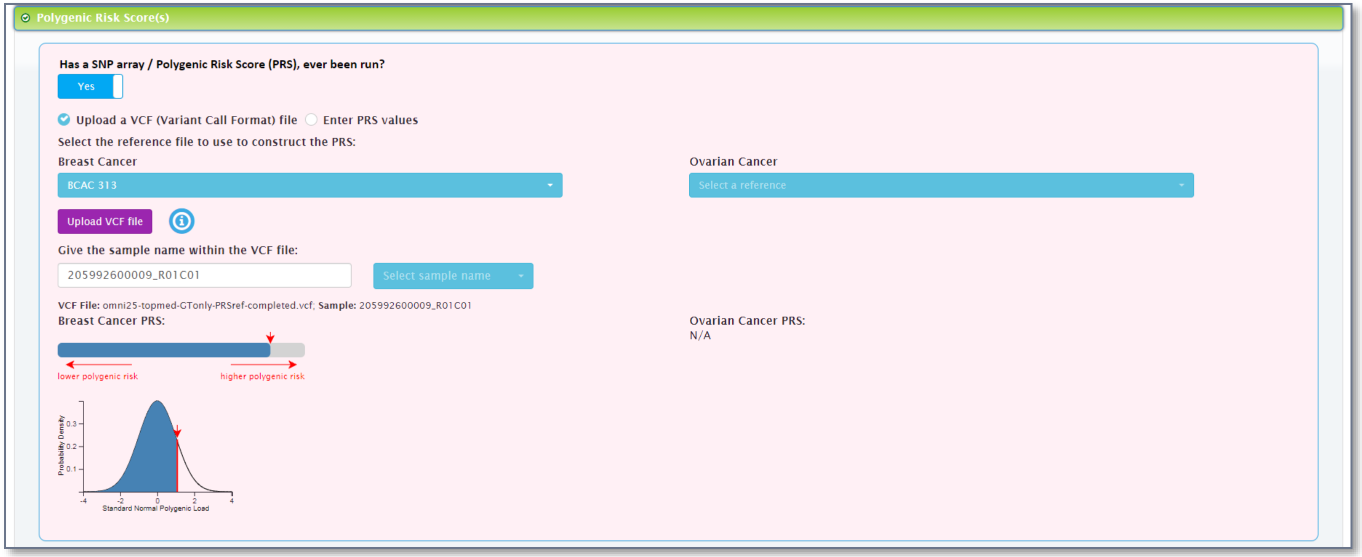
QUICK FIX FOR THE PRESENT WORK:

In order for the data to be correctly read by CanRisk, we have only uploaded the genotypes.

<table>
<tr><th>#CHROM</th><th>POS</th><th>ID</th><th>REF</th><th>ALT</th><th>QUAL</th><th>FILTER</th><th>INFO</th><th>FORMAT</th><th>205992600009_R01C01</th><th>205992600009_R02C01</th></tr>
<tr><td>1</td><td>7917076</td><td>chr1:7857016:G:A</td><td>G</td><td>A</td><td>.</td><td>PASS</td><td>AF=0.31172;MAF=0.31172;R2=0.99638;IMPUTED</td><td>GT</td><td>0|0</td><td>1|0</td></tr>
<tr><td>1</td><td>10566215</td><td>chr1:10506158:A:G</td><td>A</td><td>G</td><td>.</td><td>PASS</td><td>AF=0.38117;MAF=0.38117;R2=0.97642;ER2=0.79539;TYPED</td><td>GT</td><td>1|0</td><td>1|1</td></tr>
<tr><td>1</td><td>18807339</td><td>chr1:18480845:T:C</td><td>T</td><td>C</td><td>.</td><td>PASS</td><td>AF=0.56034;MAF=0.43966;R2=0.99141;IMPUTED</td><td>GT</td><td>0|0</td><td>1|0</td></tr>
<tr><td>1</td><td>41380440</td><td>chr1:40914768:C:T</td><td>C</td><td>T</td><td>.</td><td>PASS</td><td>AF=0.68750;MAF=0.31250;R2=1.00000;ER2=1.00000;TYPED</td><td>GT</td><td>1|1</td><td>0|0</td></tr>
</table>

POLYGENIC RISK SCORE

Once the formatting was done, the VCF file containing all the sample data could be integrated into CanRisk and the PRS score calculated



OMNI 2.5	Ref
205992600009_R01C01	Euro 1
205992600009_R02C01	AfroSub1
205992600009_R03C01	AfroSub2
205992600009_R04C01	Hisp1
205992600009_R05C01	Euro2
205992600009_R06C01	Euro3
205992600009_R07C01	Euro4
205992600009_R08C01	Euro5

Initial setting

1. The basic information integrated in the risk calculation will be as neutral as possible in order to better observe the impact of the PRS.

2. We used : female, 24 y old , no children, no other criteria filled in.

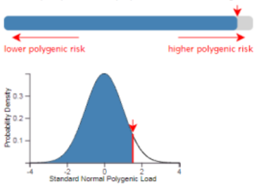
3. Various clinical risk factors will then be added and/or removed in order to better define the way they modulate the patient's risk based on PRS.

4. The different effects will then be compared with each other

POLYGENIC RISK SCORE – **Calculation**

205992600009_R02C01 – Euro1

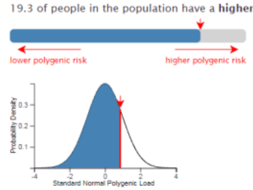
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
93.7 of people in the population have a **lower** polygenic load.
6.3 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:1.5303628288858326
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T15:40:20.726223+01:00

205992600009_R05C01 – Euro2

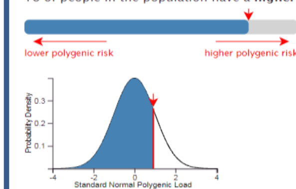
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
80.7 of people in the population have a **lower** polygenic load.
19.3 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:0.8686670096215867
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T18:29:48.164595+01:00

205992600009_R08C01 – Euro5

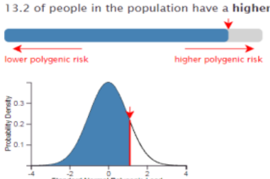
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations
Breast Cancer PRS:
82 of people in the population have a **lower** polygenic load.
18 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:0.91674813905948
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T20:40:47.580926+01:00

205992600009_R06C01 – Euro3

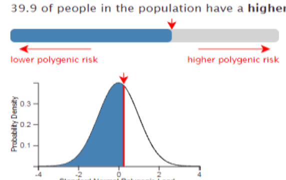
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
86.8 of people in the population have a **lower** polygenic load.
13.2 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:1.1147097547451497
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T20:43:49.360643+01:00

205992600009_R07C01- Euro4

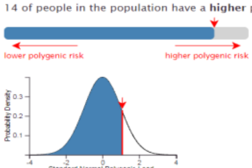
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
60.1 of people in the population have a **lower** polygenic load.
39.9 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:0.2565444927777896
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T20:36:38.172883+01:00

205992600009_R01C01 - AfroSub1

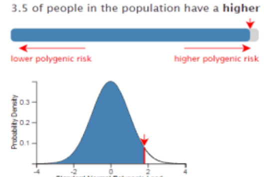
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
86 of people in the population have a **lower** polygenic load.
14 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:1.0788976031638222
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T15:44:14.425665+01:00

205992600009_R03C01- AfroSub2

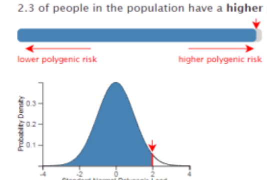
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
96.5 of people in the population have a **lower** polygenic load.
3.5 of people in the population have a **higher** polygenic load.



alpha:0.441 z-score:1.8175232295286963
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T15:47:03.784484+01:00

205992600009_R04C01- Hisp1

Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
97.7 of people in the population have a **lower** polygenic load.
2.3 of people in the population have a **higher** polygenic load.

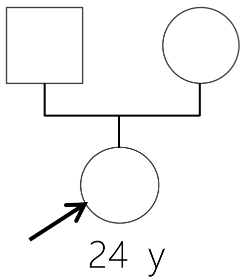


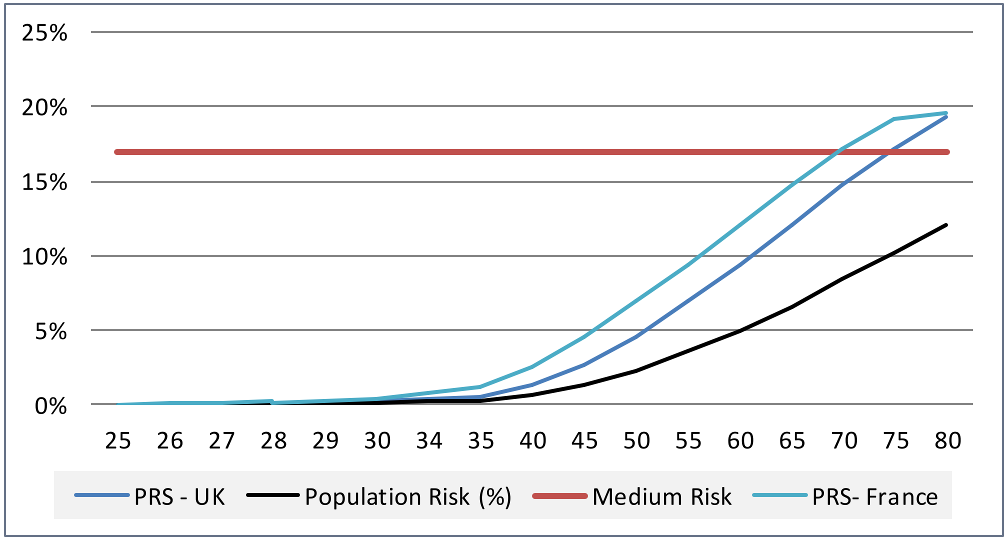
alpha:0.441 z-score:1.9932680474741002
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-20T15:58:24.821357+01:00

POLYGENIC RISK SCORE – **Lifetime Risk Calculation**

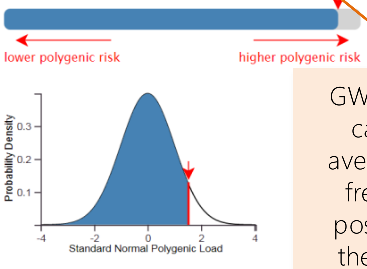
205992600009_R02C01 – Euro1

UK monogenic variants frequency by default





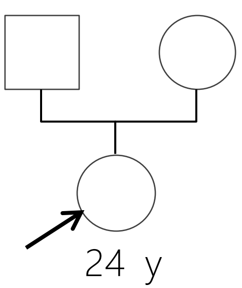
Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
93.7 of people in the population have a **lower** polygenic load.
6.3 of people in the population have a **higher** polygenic load.

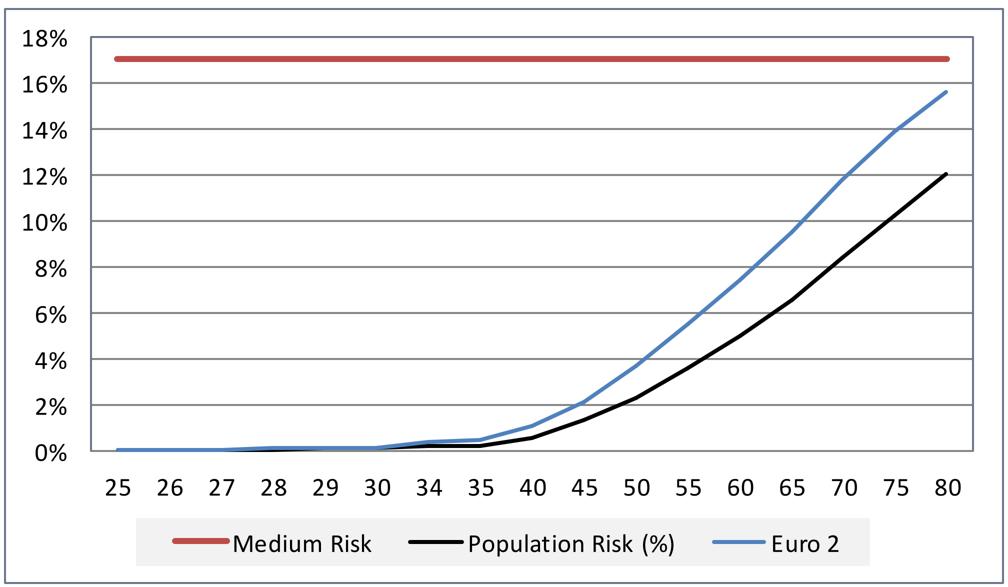


alpha:0.441 z-score:1.53036282888
Version: boadicea model 6.1.0, version 0.6
21T22:50:08.295978+01:00

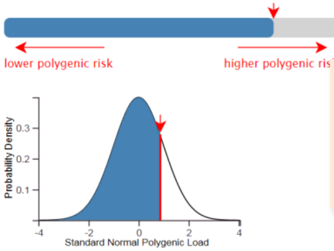
GWAS PRS model calculated on average European frequency ; not possible to adapt the PRS to other continents or specific European countries

205992600009_R05C01 – Euro2





Mutation frequency: UK
Cancer incidence rates: Netherlands
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
80.7 of people in the population have a **lower** polygenic load.
19.3 of people in the population have a **higher** polygenic load.

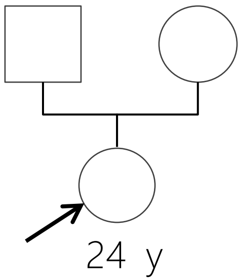


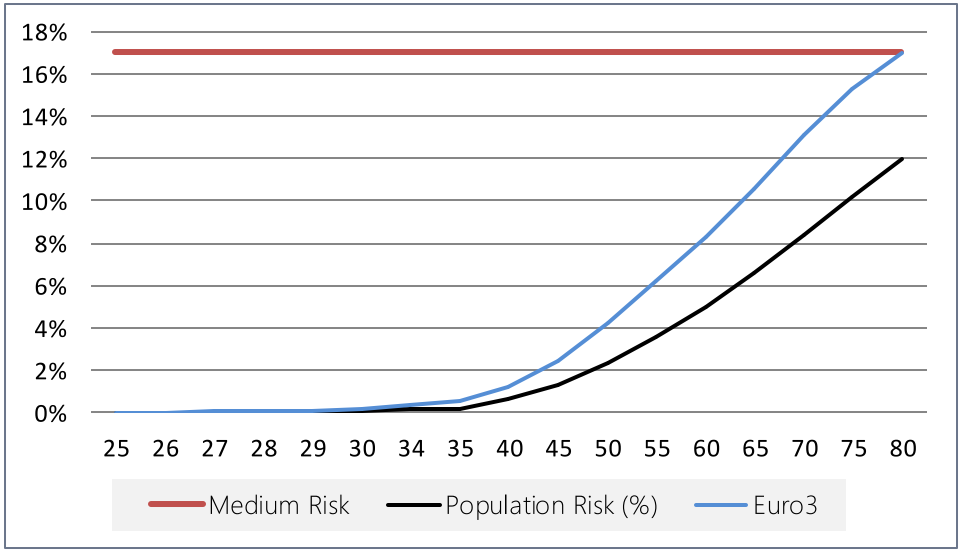
alpha:0.441 z-score:0.8686670096215867
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-21T22:54:57.432937+01:00

Not tailorable to living in Belgium cancer risk

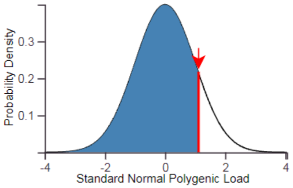
POLYGENIC RISK SCORE – Lifetime Risk Calculation

205992600009_R06C01 – Euro3



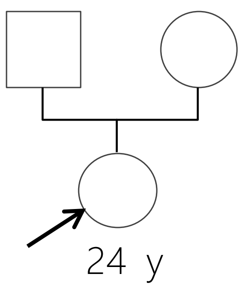


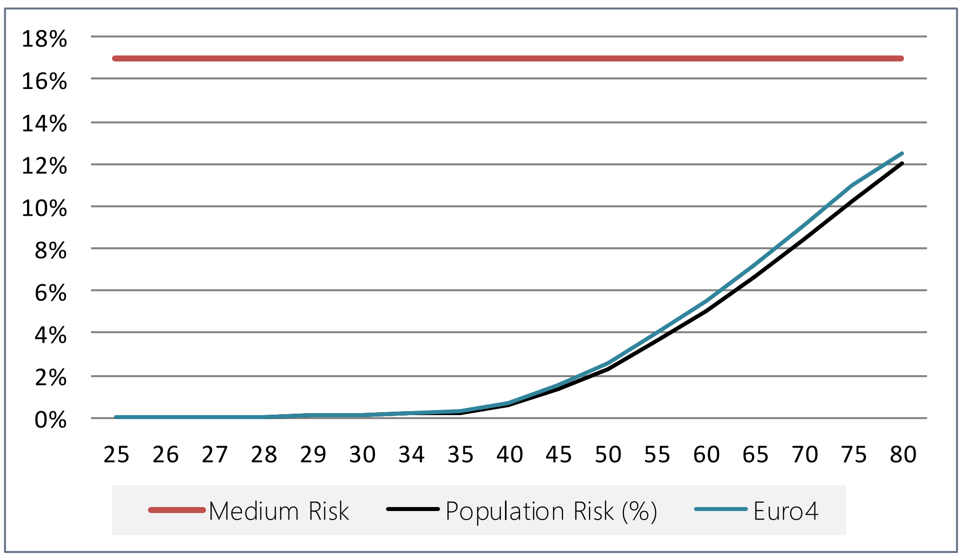
Mutation frequency: UK
Cancer incidence rates: Netherlands
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
86.8 of people in the population have a **lower** polygenic load.
13.2 of people in the population have a **higher** polygenic load.



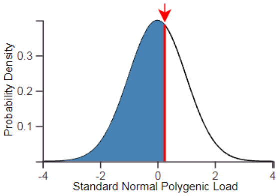
alpha:0.441 z-score:1.1147097547451497
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-21T23:08:30.037579+01:00

205992600009_R07C01- Euro4





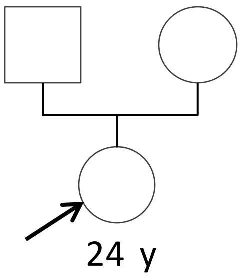
Mutation frequency: UK
Cancer incidence rates: Netherlands
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
60.1 of people in the population have a **lower** polygenic load.
39.9 of people in the population have a **higher** polygenic load.

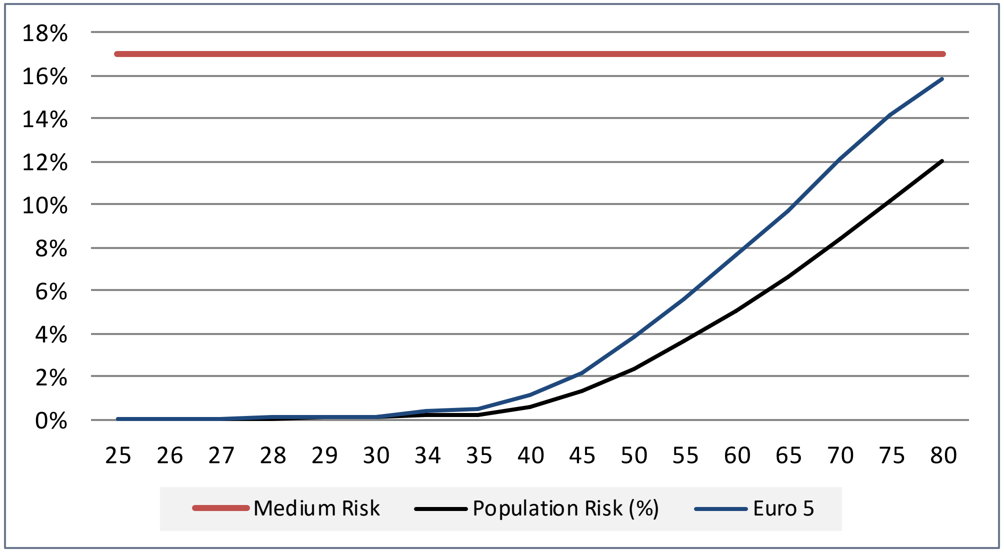


alpha:0.441 z-score:0.2565444927777896
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-21T23:12:40.499042+01:00

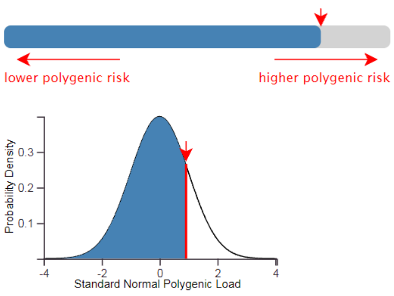
POLYGENIC RISK SCORE – Lifetime Risk Calculation

205992600009_R08C01 – Euro5





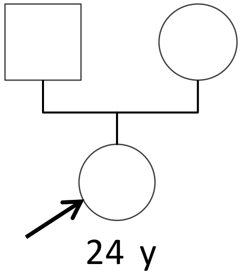
Mutation frequency: UK
Cancer incidence rates: Netherlands
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
82 of people in the population have a **lower** polygenic load.
18 of people in the population have a **higher** polygenic load.

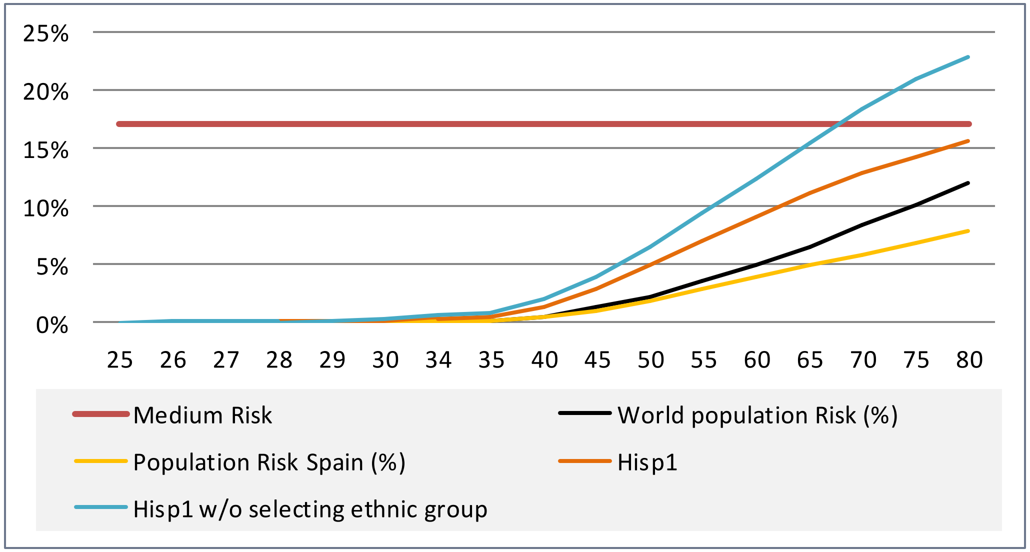


alpha:0.441 z-score:0.91674813905948
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-21T23:17:37.327891+01:00

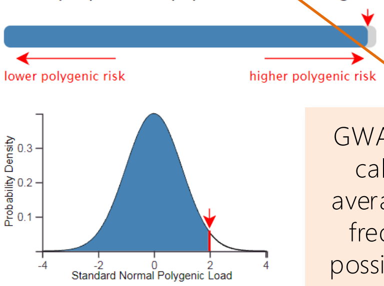
205992600009_R04C01- Hisp1

Tailorable to living in Spain cancer rate risk





Mutation frequency: UK
Cancer incidence rates: Spain
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
97.7 of people in the population have a **lower** polygenic load.
2.3 of people in the population have a **higher** polygenic load.

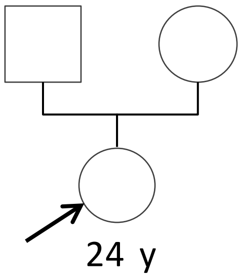


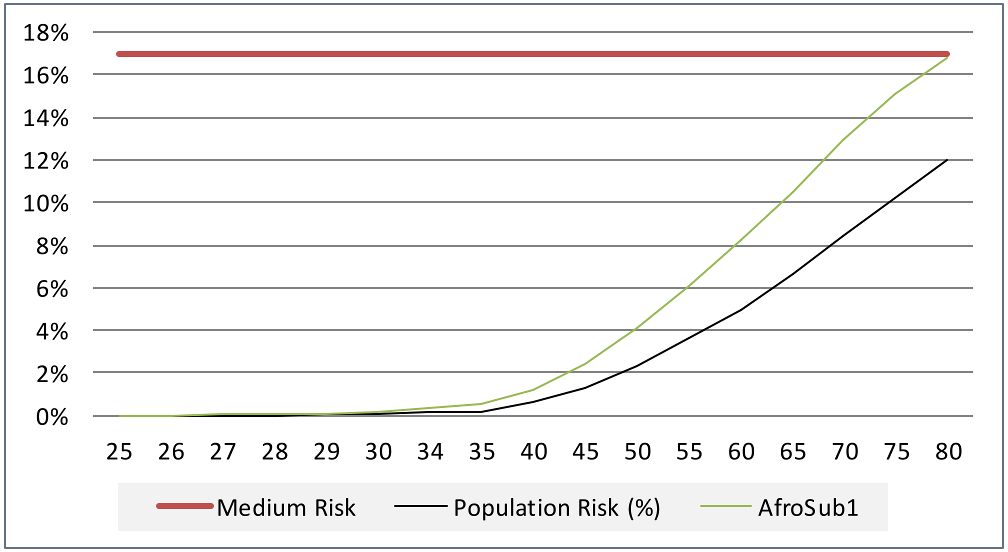
alpha:0.441 z-score:1.99326804747
Version: boadicea model 6.1.0, version 0.6
22T17:08:37.269910+01:00

GWAS PRS model calculated on average European frequency ; not possible to choose spain ethnicity for PRS model

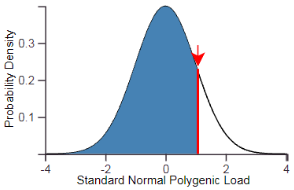
POLYGENIC RISK SCORE – Lifetime Risk Calculation

205992600009_R01C01 - AfroSub1





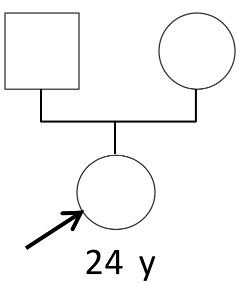
Mutation frequency: UK
Cancer incidence rates: Netherlands
Please note the model has been developed using data from European ancestry populations.
Breast Cancer PRS:
86 of people in the population have a **lower** polygenic load.
14 of people in the population have a **higher** polygenic load.

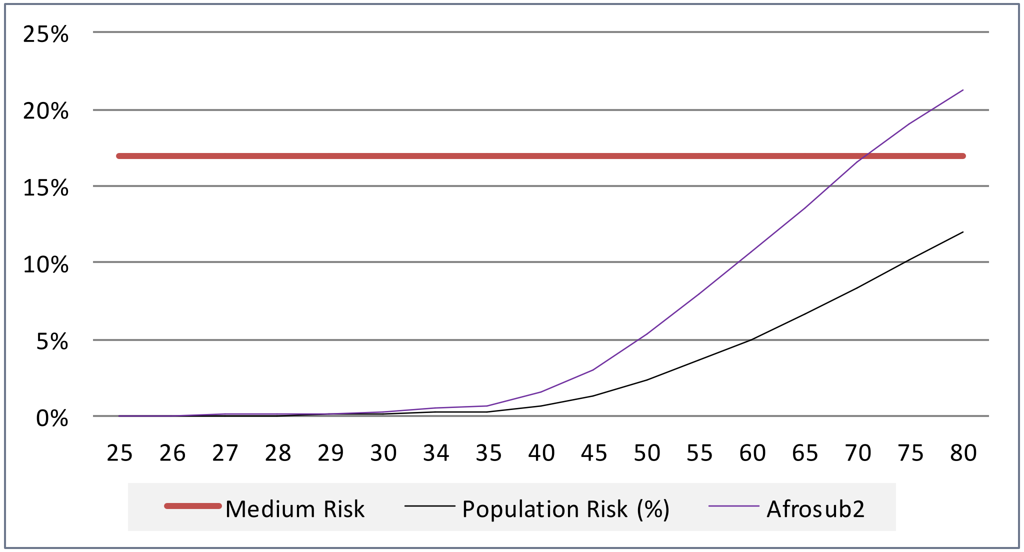


alpha:0.441 z-score:1.0788976031638222
Version: boadicea model 6.1.0, version 0.
21T23:34:15.049239+01:00

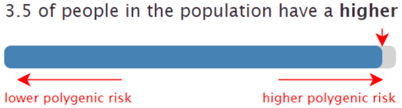
Living country and GWAS PRS model not tailorable to African countries or ethnicities

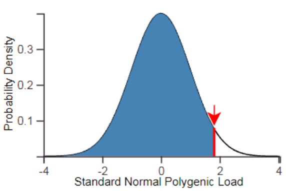
205992600009_R03C01- AfroSub2





Mutation frequency: UK
Cancer incidence rates: UK
Please note the model has been developed using data fron European ancestry populations.
Breast Cancer PRS:
96.5 of people in the population have a **lower** polygenic load.
3.5 of people in the population have a **higher** polygenic load.

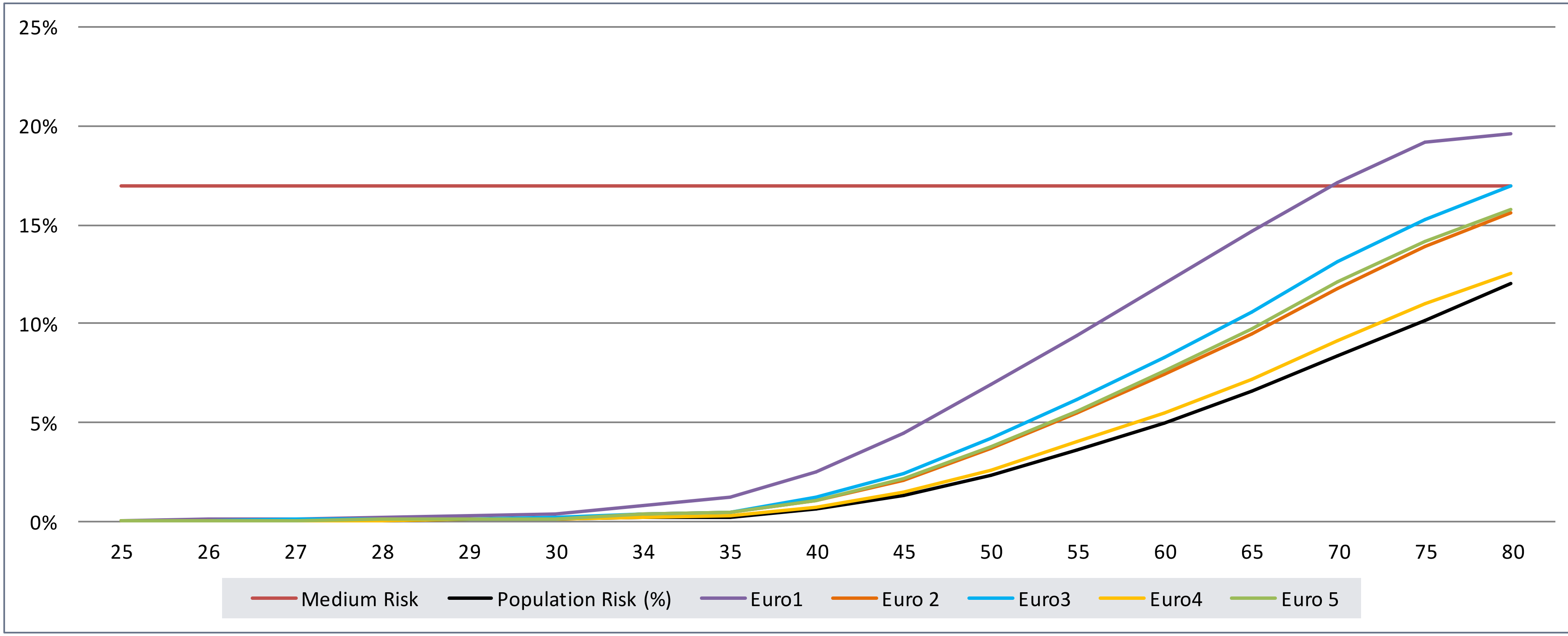




alpha:0.441 z-score:1.8175232295286963
Version: boadicea model 6.1.0, version 0.6.0; **Timestamp:** 2022-05-21T23:39:24.763934+01:00

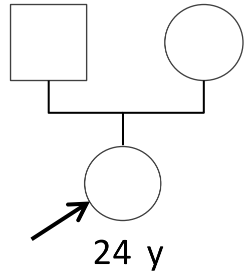
POLYGENIC RISK SCORE – **Lifetime Risk Calculation**

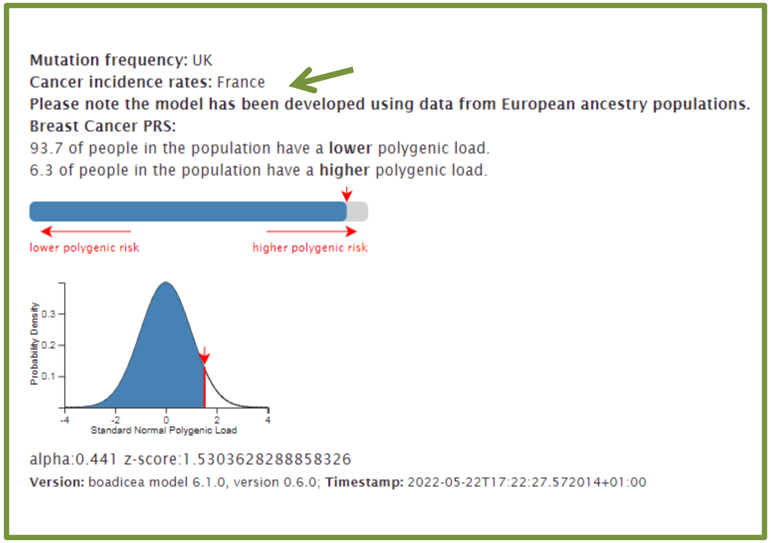
PRS alone could impact the clinical follow-up of 2 European patients (! 53 missing SNPs)

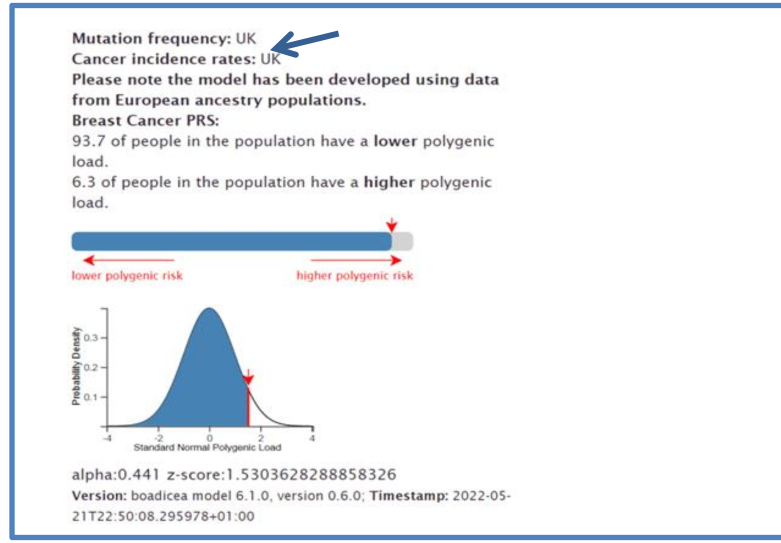


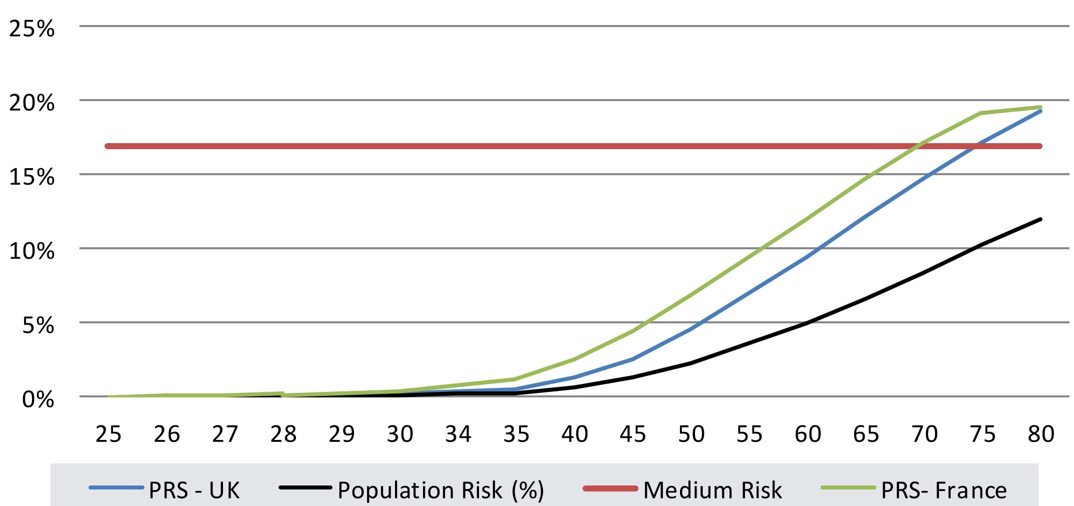
age	25	26	27	28	29	30	34	35	40	45	50	55	60	65	70	75
Medium Risk	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%
Population Risk (%)	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**1%**	**1%**	**2%**	**4%**	**5%**	**7%**	**8%**	**10%**
Euro1	0,0%	0,1%	0,1%	0,2%	0,3%	0,4%	0,8%	1,2%	2,5%	4,5%	6,9%	9,4%	12,0%	14,7%	17,1%	19,2%
Euro 2	0%	0,0%	0,0%	0,1%	0,1%	0,1%	0,4%	0,5%	1,1%	2,1%	3,7%	5,5%	7,4%	9,5%	11,8%	13,9%
Euro3	0%	0,0%	0,1%	0,1%	0,1%	0,2%	0,4%	0,5%	1,2%	2,4%	4,2%	6,2%	8,3%	10,6%	13,1%	15,3%
Euro4	0%	0,0%	0,0%	0,0%	0,1%	0,1%	0,2%	0,3%	0,7%	1,5%	2,6%	4,0%	5,5%	7,2%	9,1%	11,0%
Euro 5	0%	0,0%	0,0%	0,1%	0,1%	0,1%	0,4%	0,5%	1,1%	2,2%	3,8%	5,6%	7,6%	9,7%	12,1%	14,2%

TESTS FOR CLINICAL FACTORS IMPACT ON PRS LIFETIME RISK



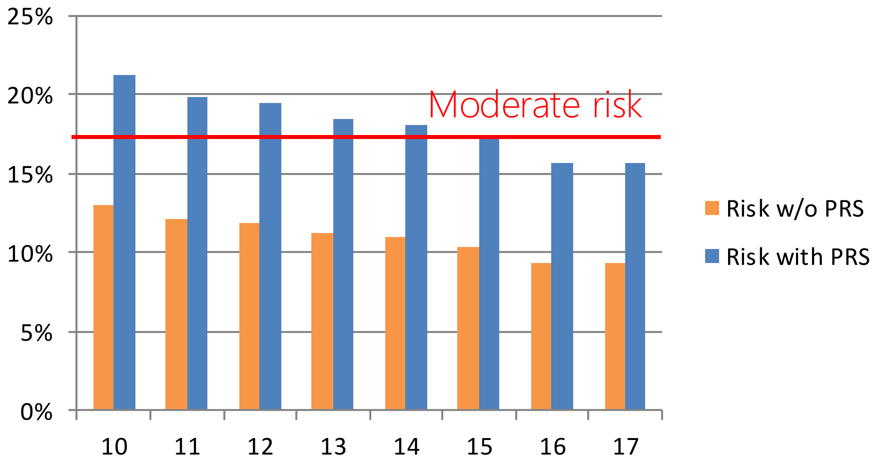




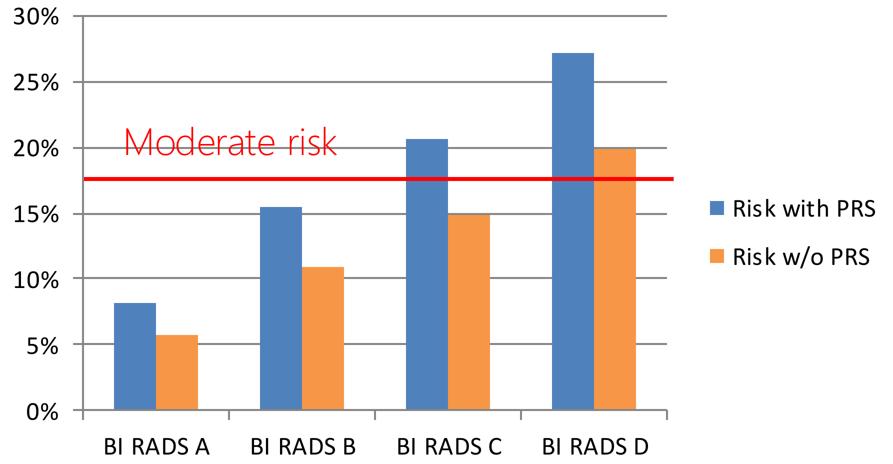


TESTS FOR CLINICAL FACTORS IMPACT ON EURO 1 PRS LIFETIME RISK

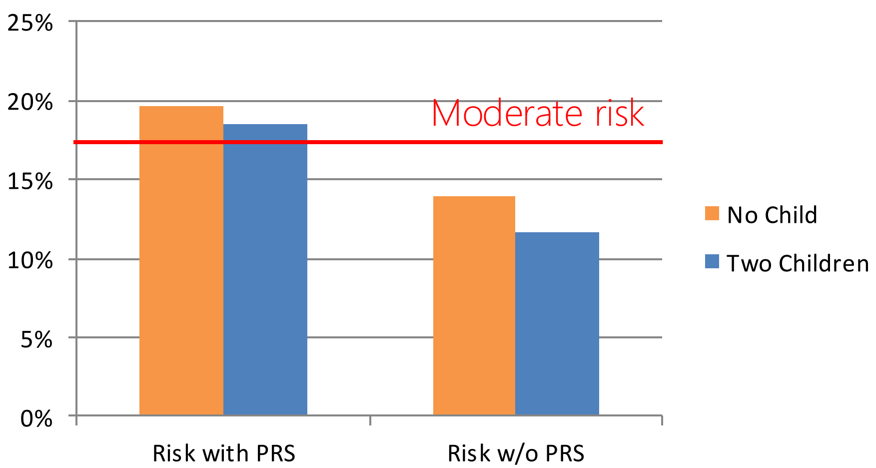
Age of menarche



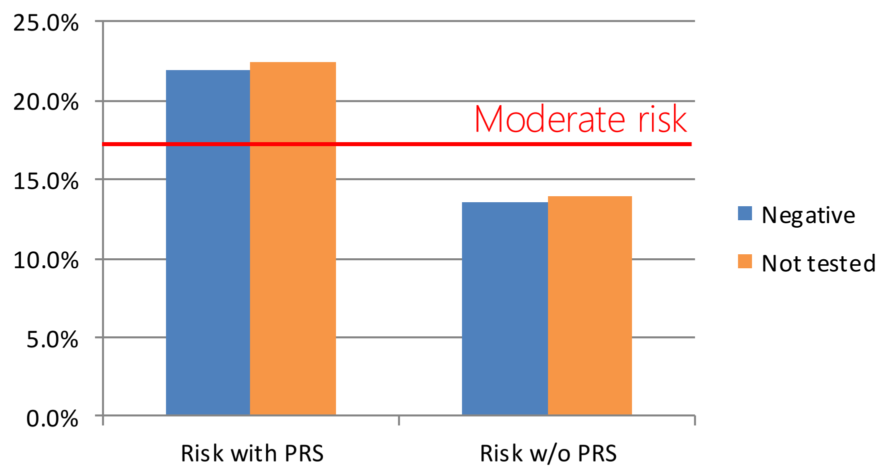
Breast density



Children



Gene panel



POLYGENIC RISK SCORE – TO DO LIST

<u>Technical Improvements</u>

- Analysis of the OMNI 5 data (better than the 2.5 ?)

- Missing SNPs : impute with Michigan server and Sanger server (will other reference panels and imputation algorithms retrieve some missing SNPs ?)

- PRS improvement :
  - PRS of 333 SNPs is now available, but is not yet integrated to CanRisk PMID: 32424353
  - GWAS PRS model calculation would be available for any ethnicity.
  - cancer incidence rate would be available for every country in the world.

- Automate imputation and file formatting.

- Develop quality control criteria and produce an automated QC report

- Design a prescription with the minimal needed clinical information for PRS

- Automate CanRisk uploading and downloading

POLYGENIC RISK SCORE – PERSPECTIVES

<u>Clinical thoughts</u>

- Define inclusion criteria for PRS tests (age, personal and familial history)

- Are there cases where it would be appropriate to perform a PRS score in 1st intention ?

- Evaluate familial segregation of PRS in order to develop genetic counseling habits

- Can we use reliably the BCAC 313 PRS for any European patient (North – South – West – East - Europeans, Benelux, Nordic, Hispanic, Mediterranean , etc.) ?

- Which cancer incidence rates must be chosen for migrants : country of original residence or country of actual residence ? For example, a Spanish Citizen having lived 25y in Spain and now living for 10y in Belgium.

- Must we refrain ourselves using the BCAC 313 PRS test for non European patients ?

- Is it wise (sufficiently clinically validated) to start using the BCAC 313 PRS score on European patients ?

ADDITIONAL SLIDES

BREAST CANCER – RISK FACTORS

Breast cancer is a complex and heterogeneous disease.

Systemic health features

- **Age:** Breast cancer risk increases after age 20
- **Age at menarche** : The onset of menses prior to age 13 has been associated with an increased risk for breast cancer, while onset after age 15 has been shown to decrease risk (a 10% reduction in lifetime breast cancer risk per 2 years).
- **Age at menopause :** each 5-year delay in menopause has been associated with a ~17% increase in lifetime breast cancer risk
- **Breast density**
- **History of endometriosis**
- History of cancer : (bilateral) breast (excluding DCIS),ovarian cancer, and pancreatic

Environmental exposures

- History of HRT use, including duration of use, use within the last 5 years, and type of HRT used
- History of birth control pill use

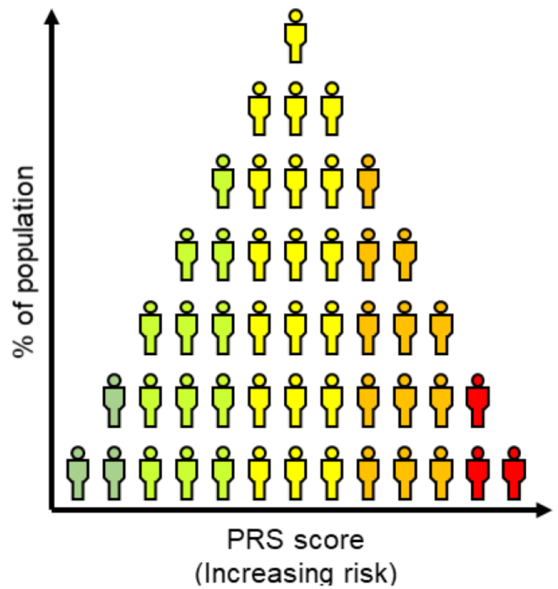
Lifestyle

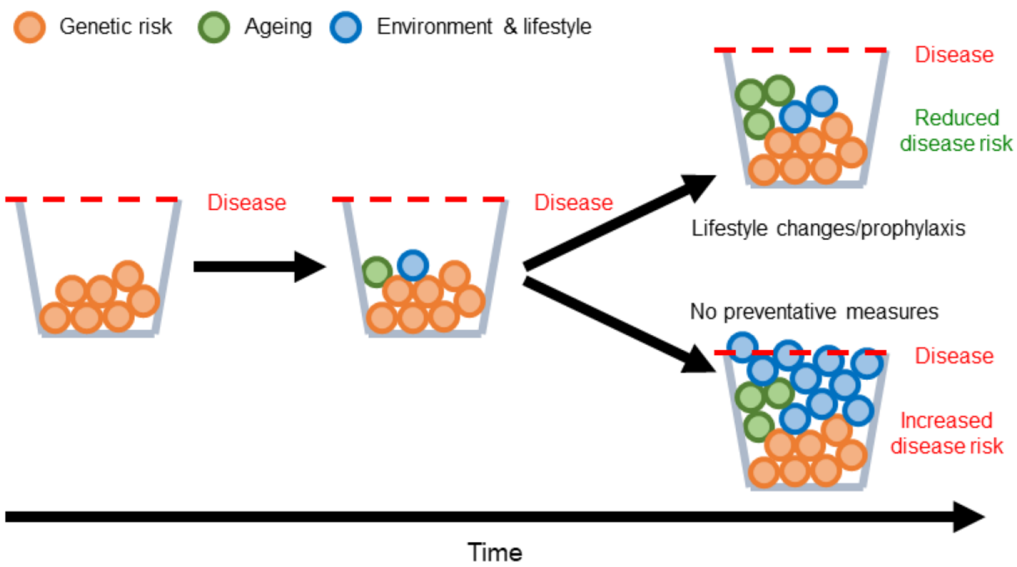
- Age at first live birth : Women with a first live birth at or before age 20 show a ~20% reduction in lifetime breast cancer risk
- History of surgical procedures, including tubal liga-tion, oophorectomy, and mastectomy
- Alcohol consumption
- tobacco use

Genetic aspects

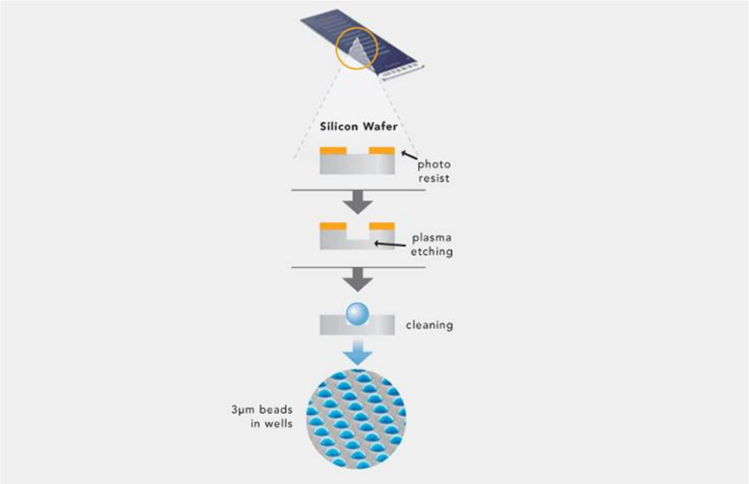
- Polygenic risk score
- Ethnicities :
  - higher rates among Caucasian women of European ancestry
  - Black women, are more likely to be diagnosed with aggressive, triple-negative breast cancer, at advanced stages and at younger ages,
- Ashkenazi Jewish ancestry
- Family history including genetic testing and age of diagnosis of cancer breast (male and female), ovarian ,prostate and pancreatic

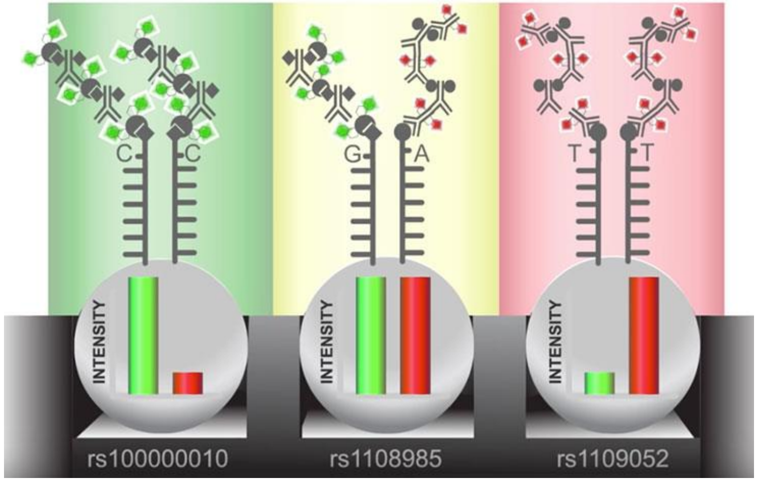
COMPLEX GENETICS



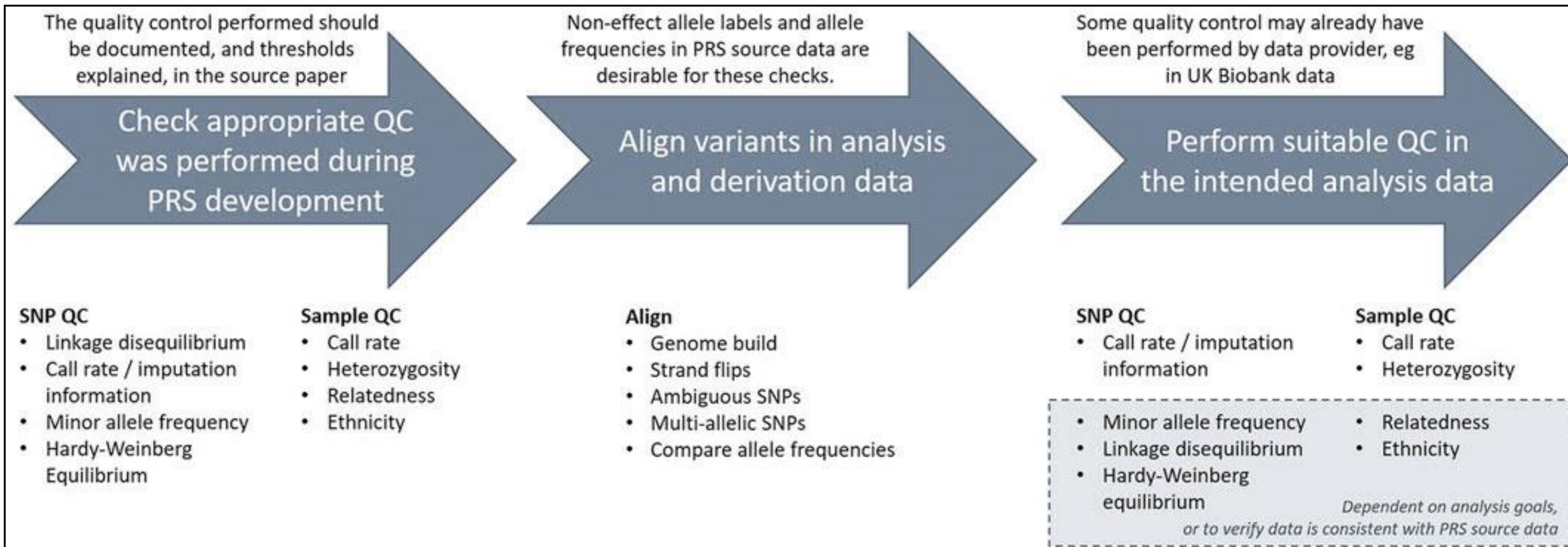


ILLUMINA SNP ARRAY HYBRIDATION





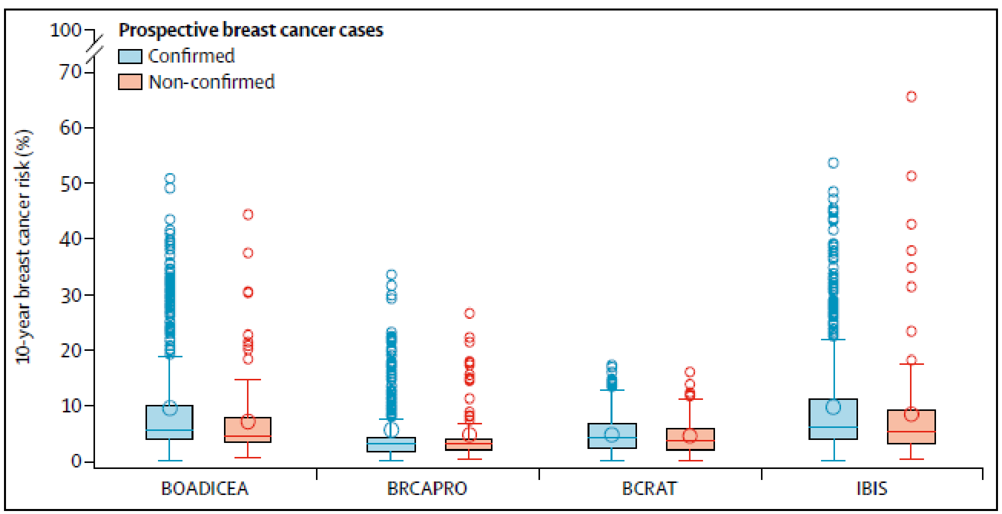
SNP ARRAY AND IMPUTATION QUALITY CONTROL



CANRISK TOOL PERFORMANCE

Risk factors and limitations of the CanRisk model.

	Personalized risk factors	Limitations
CanRisk	• Current age • BMI • Alcohol consumption • Age at menarche • Age at menopause • Age at first live birth • History of birth control pill use • History of HRT use, including duration of use, use within the last 5 years, and type of HRT used • Breast density • History of endometriosis • History of surgical procedures, including tubal ligation, oophorectomy, and mastectomy • History of (bilateral) breast cancer (excluding DCIS), ovarian cancer, and pancreatic cancer • Polygenic risk score • Ashkenazi Jewish ancestry • Family history including genetic testing and age of diagnosis of breast cancer (male and female), ovarian cancer or prostate cancer, and pancreatic cancer	• Family history is limited to only first-degree and some second-degree relatives (does not include cousins). • Cannot be used for women with more than 2 breast cancers. • Does not consider breast cancer risk from mantle radiation • Model validation has only been completed within the UK, limiting generalizability to individuals of non-European ancestries or ethnicities • Does not account for competing mortality



	Total breast cancer cases	Sensitivity (95% CI)	Specificity (95% CI)
Overall population			
N	619	..	..
BOADICEA	..	81·9% (78·6–84·9)	42·8% (42·1–43·6)
≥3·4%	507	..	..
<3·4%	112	..	..
BRCAPRO	..	45·9% (41·9–49·9)	76·1% (75·4–76·8)
≥3·4%	284	..	..
<3·4%	335	..	..
BCRAT	..	57·8% (53·8–61·8)	55·0% (54·2–55·8)
≥3·4%	358	..	..
<3·4%	261	..	..
IBIS	..	79·5% (76·1–82·6)	46·7% (45·9–47·5)
≥3·4%	492	..	..
<3·4%	127	..	..

	Expected number of breast cancer cases	Observed number of breast cancer cases	Expected/observed ratio (95% CI)	Concordance statistic (95% CI)
Overall population (n=15 732)				
BOADICEA	652·5	619	1·05 (0·97–1·14)	0·70 (0·68–0·72)
BRCAPRO	367·7	619	0·59 (0·55–0·64)	0·68 (0·65–0·70)
BCRAT	488·6	619	0·79 (0·73–0·85)	0·60 (0·58–0·62)
IBIS	640·3	619	1·03 (0·96–1·12)	0·71 (0·69–0·73)
***BRCA* negative (n=14 657)**				
BOADICEA	492·3	482	1·02 (0·93–1·12)	0·65 (0·62–0·67)
BRCAPRO	257·6	482	0·53 (0·49–0·58)	0·62 (0·59–0·64)
BCRAT	466·7	482	0·97 (0·89–1·06)	0·64 (0·61–0·66)
IBIS	484·1	482	1·00 (0·92–1·10)	0·66 (0·64–0·68)
***BRCA* positive (n=1075)**				
BOADICEA	160·2	137	1·17 (0·99–1·38)	0·59 (0·53–0·64)
BRCAPRO	110·1	137	0·80 (0·68–0·95)	0·57 (0·52–0·63)
BCRAT	..	..	..	..
IBIS	156·2	137	1·14 (0·96–1·35)	0·60 (0·55–0·66)
Age <50 years at baseline (n=9798)				
BOADICEA	331·5	330	1·00 (0·90–1·12)	0·75 (0·73–0·78)
BRCAPRO	181·0	330	0·55 (0·49–0·61)	0·73 (0·70–0·76)
BCRAT	181·7	330	0·55 (0·49–0·61)	0·60 (0·57–0·63)
IBIS	328·7	330	1·00 (0·89–1·11)	0·75 (0·72–0·78)
Age <50 years at baseline and *BRCA* negative* (n=8397)				
BOADICEA	221·0	218	1·01 (0·89–1·16)	0·67 (0·64–0·70)
BRCAPRO	97·5	218	0·45 (0·39–0·51)	0·63 (0·59–0·66)
BCRAT	169·3	218	0·78 (0·68–0·89)	0·64 (0·61–0·67)
IBIS	209·6	218	0·96 (0·84–1·10)	0·67 (0·63–0·70)

All eligible participants in the Breast Cancer Prospective Family Study Cohort are shown for each category. BOADICEA=Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm model. BCRAT=Breast Cancer Risk Assessment Tool. IBIS=International Breast Cancer Intervention Study model. *Women not known to be *BRCA1* or *BRCA2* carriers (includes both tested and untested women).

CHOOSING A POLYGENIC RISK SCORE

The PGS Catalog Project

The PGS Catalog is an open database of published polygenic scores. Each PGS in the Catalog is consistently annotated with relevant metadata; including scoring files (variants, effect alleles/weights).

https://www.pgscatalog.org/about/

The Polygenic Score Catalog as an open database for reproducibility and systematic evaluation
Nature Genetics
doi: 10.1038/s41588-021-00783-5 (2021).

The Polygenic Score Catalog as an open database for reproducibility and systematic evaluation

We present the Polygenic Score (PGS) Catalog (https://www.PGSCatalog.org), an open resource of published scores (including variants, alleles and weights) and consistently curated metadata required for reproducibility and independent applications. The PGS Catalog has capabilities for user deposition, expert curation and programmatic access, thus providing the community with a platform for PGS dissemination, research and translation.

Samuel A. Lambert, Laurent Gil, Simon Jupp, Scott C. Ritchie, Yu Xu, Annalisa Buniello, Aoife McMahon, Gad Abraham, Michael Chapman, Helen Parkinson, John Danesh, Jacqueline A. L. MacArthur and Michael Inouye

Cancer PRSweb

Freely available genome-wide association studies (GWAS) summary statistics from multiple sources.

It is an online repository for polygenic risk scores (PRS) for common cancer traits. For each PRS construct, measures on performance, calibration, and discrimination are provided.

https://prsweb.sph.umich.edu:8443/

Cancer PRSweb: An Online Repository with Polygenic Risk Scores for Major Cancer Traits and Their Evaluation in Two Independent Biobanks,
The American Journal of Human Genetics
doi.org/10.1016/j.ajhg.2020.08.025 (2020).

Cancer PRSweb: An Online Repository with Polygenic Risk Scores for Major Cancer Traits and Their Evaluation in Two Independent Biobanks

Lars G. Fritsche,[1,2,3,7,8,*] Snehal Patil,[1,2] Lauren J. Beesley,[1,3] Peter VandeHaar,[1,2] Maxwell Salvatore,[1,3] Ying Ma,[1,2] Robert B. Peng,[1,3,4] Daniel Taliun,[1,2] Xiang Zhou,[1,2,3] and Bhramar Mukherjee[1,2,3,5,6,7,*]

Summary

To facilitate scientific collaboration on polygenic risk scores (PRSs) research, we created an extensive PRS online repository for 35 common cancer traits integrating freely available genome-wide association studies (GWASs) summary statistics from three sources: published GWASs, the NHGRI-EBI GWAS Catalog, and UK Biobank-based GWASs. Our framework condenses these summary statistics into PRSs using various approaches such as linkage disequilibrium pruning/p value thresholding (fixed or data-adaptively optimized thresholds) and penalized, genome-wide effect size weighting. We evaluated the PRSs in two biobanks: the Michigan Genomics Initiative (MGI), a longitudinal biorepository effort at Michigan Medicine, and the population-based UK Biobank (UKB). For each PRS construct, we provide measures on predictive performance and discrimination. Besides PRS evaluation, the Cancer-PRSweb platform features construct downloads and phenome-wide PRS association study results (PRS-PheWAS) for predictive PRSs. We expect this integrated platform to accelerate PRS-related cancer research.

Standard equation to calculate a weighted polygenic risk score

Calculating the polygenic risk score for a patient → PRS_j

SNP = corresponds to a change of one nucleotide → i

We define a set of SNPs involved in a pathology → N

Each SNP is assigned an effect size → β_i

obtained from a genome-wide association study (GWAS)

the impact of a SNP in a patient depends on the effect size and copy number of that SNP in the patient → $\beta_i * dosage_{ij}$

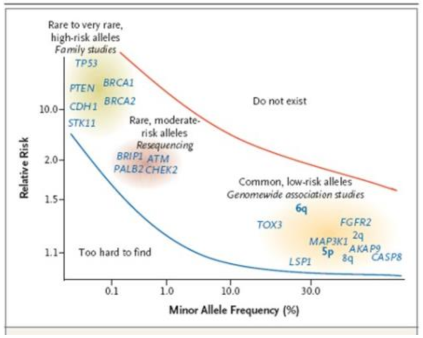
Repeating all the operation for each SNP and summing the data for all SNPs gives a cumulative risk of risk

$$PRS_j = \sum_{i}^{N} \beta_i * dosage_{ij}$$

VCF GENOTYPING INFORMATION

Allele Frequency

- MAF: The minor allele frequency indicates how rare a variant is.
- Reference allele : the allele which is most common in the population is known as the “major”
- Alternative allele: allele and the less common allele(s) are “minor.”



Calculating Dosages

GT:DS:HDS:GP - **A** allèle *ref* = 0 0|0:0:0,0:1,0,0 - **B** *allèle alt* = 1	**AA**	**AB**	**BA**	**BB**
GT - Estimated most likely genotype	0 | 0	0 | 1	1| 0	1 | 1
DS - Estimated alternate allele dosage [0*P(0/0) + P(0/1)+2*P(1/1)] P (AA) **0** **2 P(BB)** Hard-called, or thresholded, dosages are integer values, $dosage_{ij}$ ∈{ 0 , 1 , 2} for SNP i in individual j, that are obtained by choosing a threshold value at which to round the expected (allelic) dosage to a whole number.	0	1	1	2
HDS - Estimated phased haploid alternate allele dosage: P(B) , P(B)	0 , 0	0 , 1	1 , 0	1 , 1
GP - Estimated Posterior Genotype Probabilities P(0/0), P(0/1) and P(1/1) = P(AA) , P(AB) , P(BB)	1 , 0 , 0	0 , 1 , 0	0 , 1 , 0	0 , 0 , 1

EXTRACTING SNPs

Software Considerations

There are a range of software tools that can be used to read and manipulate this data.

In this document the calculation of the PRS is in PLINK 2.

- first auto-convert the entire data file to PLINK 2 binary format (.pgen, .pvar, .psam).
- PLINK 2 collapses the trios of raw genotype probabilities into single dosages

PLINK 2

Function	Summary statistics	As exclusion criteria	
		Option	Meaning
Allele frequency	--freq	--maf [threshold]	Include SNPs with MAF above [threshold] (default = 0.01)
SNP call rate	--missing	--geno [threshold]	Exclude SNPs with missing call rates exceeding the [threshold] (default = 0.1)
Filter SNPs		--exclude [file]	Exclude SNPs listed in [file]
Filter samples		--keep [file]	Retains only the samples listed in [file], all others are excluded
HWE	--hardy	--hwe [threshold]	Exclude SNPs with *p*-values below [threshold]
Linkage Disequilibrium (LD)	--r2*	--indep-pairwise [window][step][threshold]	Pruning with a [window] size, sliding across the genome with [step] size at a time and filter out any SNPs with LD r^2 higher than [threshold]

*Command in PLINK 1.9.

Aligning SNPs Between Base and Validation Data

It is important to ensure that the correct variants have been identified for inclusion in the PRS.

```
Target Sequence
5'  ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
          |||| |||||| ||||||||||||||
Query Sequence  5'  TACTCACGGATGAGGTACTTTAGAGGC 3'
```

Aligning / imputation issues

Strand-Flipping And Ambiguous SNPs

		PRS summary data file		Validation data	
		Effect allele	Non-effect allele	Effect allele	Non-effect allele
1	Expected scenario - perfect agreement	A	C	A	C
2	PRS and validation data disagree on labelling of effect allele	A	C	C	A
3	"Strand flip"	A	C	T	G
4	Strand flip and labelling disagreement	A	C	G	T
5	Palindromic	A	T	T	A

Multi-Allelic SNPs

Multi-allelic SNPs have multiple possible alternate alleles for one reference allele, and these can be represented and identified in different ways in different data formats. Sometimes they sharing the same rsID and chromosome position and with the same listed reference allele but different alternate alleles.

Compare Allele Frequencies

It is important to compare the frequencies of these alleles in the validation data. The population can contain groups of individuals with different ancestries.

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At a Glance

Estimated New Cases in 2023	297,790
% of All New Cancer Cases	15.2%

Estimated Deaths in 2023	43,170
% of All Cancer Deaths	7.1%

5-Year Relative Survival

90.8%

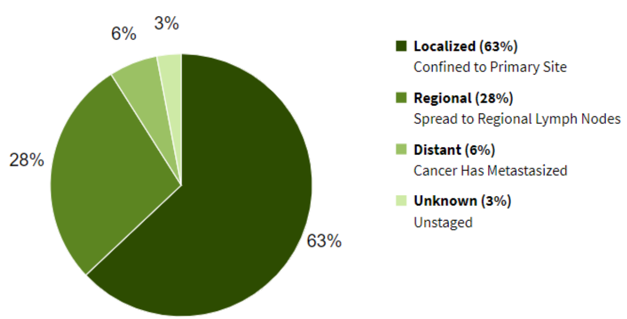
2013–2019

5-Year Relative Survival

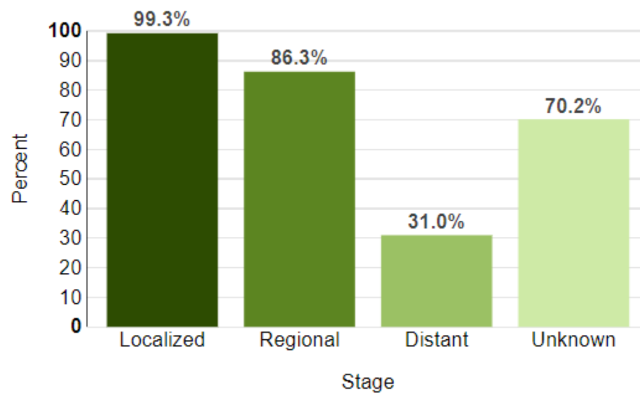
90.8%

Percent of Cases & 5-Year Relative Survival by Stage at Diagnosis: Female Breast Cancer

Percent of Cases by Stage



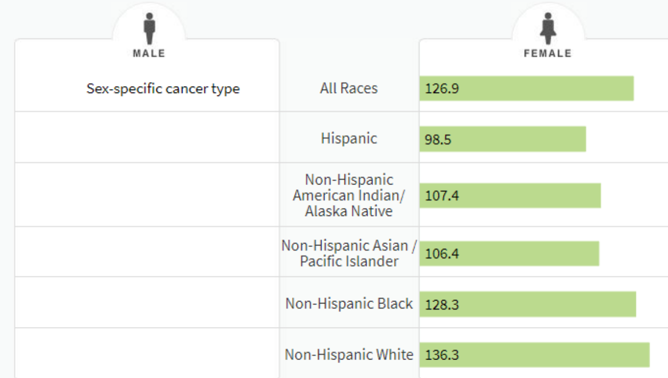
5-Year Relative Survival



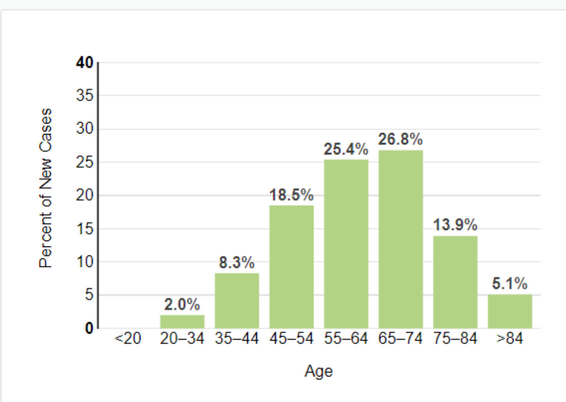
	Common Types of Cancer	Estimated New Cases 2023	Estimated Deaths 2023
1.	**Breast Cancer (Female)**	**297,790**	**43,170**
2.	Prostate Cancer	288,300	34,700
3.	Lung and Bronchus Cancer	238,340	127,070
4.	Colorectal Cancer	153,020	52,550
5.	Melanoma of the Skin	97,610	7,990
6.	Bladder Cancer	82,290	16,710
7.	Kidney and Renal Pelvis Cancer	81,800	14,890
8.	Non-Hodgkin Lymphoma	80,550	20,180
9.	Uterine Cancer	66,200	13,030
10.	Pancreatic Cancer	64,050	50,550

Female breast cancer represents 15.2% of all new cancer cases in the U.S.

Rate of New Cases per 100,000 Persons by Race/Ethnicity: Female Breast Cancer



Percent of New Cases by Age Group: Female Breast Cancer

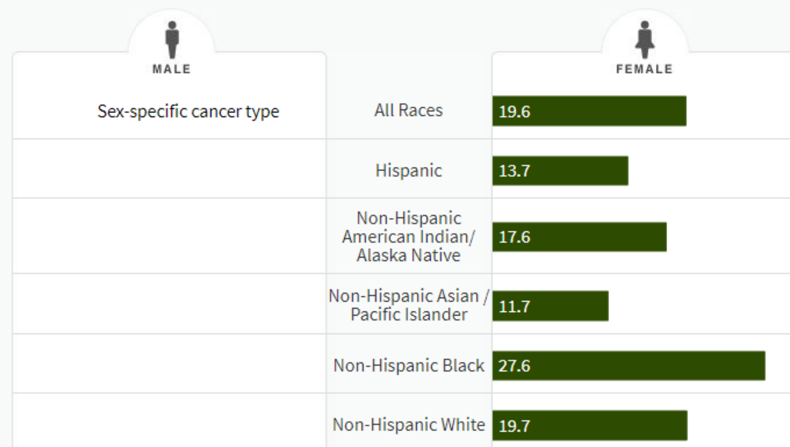


Female breast cancer is most frequently diagnosed among women aged 65–74.

Median Age At Diagnosis

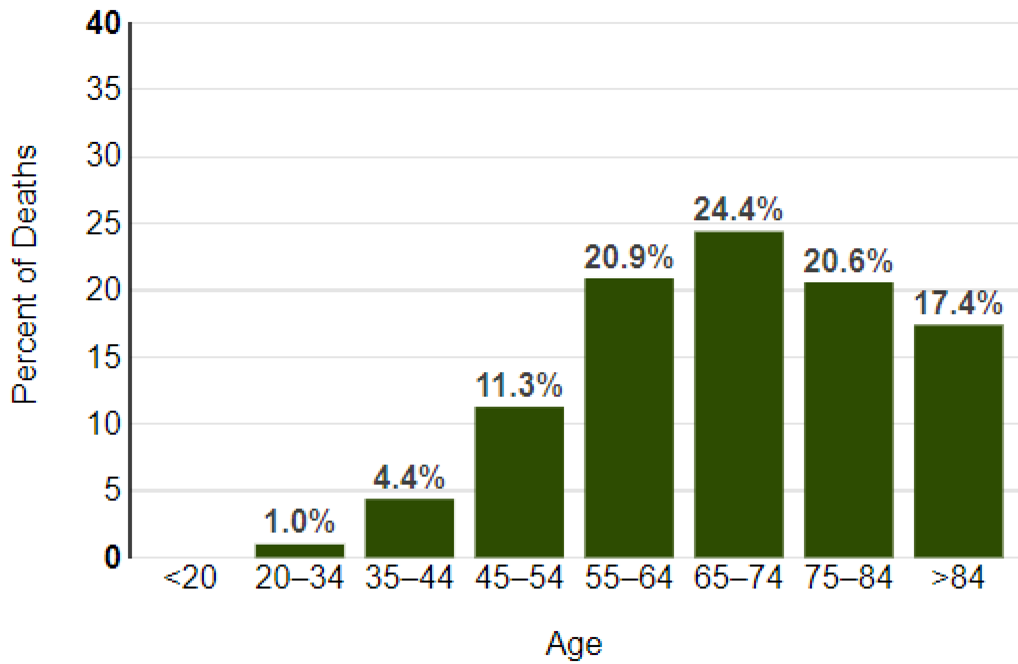
63

Death Rate per 100,000 Persons by Race/Ethnicity: Female Breast Cancer



https://seer.cancer.gov/statfacts/html/breast.html

Percent of Deaths by Age Group: Female Breast Cancer



The percent of female breast cancer deaths is highest among women aged 65–74.

Median Age At Death

70

69 études

94,075 cas de cancer du sein

75,017 sujets témoins

ensemble d'entraînement

C'est l'ensemble de données sur lequel le modèle prédictif (dans ce cas, le Score de Risque Polygénique ou PRS) est construit. Les données de cet ensemble sont utilisées pour "entraîner" le modèle à comprendre les relations entre les différents facteurs (par exemple, les variations génétiques spécifiques) et l'issue que nous cherchons à prédire (dans ce cas, le développement d'un cancer du sein).

11,428 cas de cancer du sein invasif et 18,323 sujets témoins

ensemble de validation

Une fois que le modèle est entraîné, nous devons vérifier à quel point il est précis pour prédire des résultats sur de nouvelles données qu'il n'a jamais vues auparavant. C'est là qu'intervient l'ensemble de validation. Il est séparé de l'ensemble d'entraînement et n'est utilisé que pour tester l'exactitude du modèle. Si le modèle prédit avec précision les résultats sur l'ensemble de validation, nous pouvons être plus confiants dans son utilité et sa fiabilité.

190,040 femmes de l'UK Biobank, qui n'avaient eu aucun diagnostic de cancer ou de mastectomie avant leur recrutement. Parmi ces femmes, 3,215 ont développé un cancer du sein invasif au cours d'un suivi prospectif.

Génotypage

iCOGS et OncoArray sont deux types de puces à ADN utilisées pour le génotypage. Les deux sont conçus pour étudier les variations génétiques associées au cancer, mais ils diffèrent par les SNPs spécifiques qu'ils peuvent détecter.

Puce de génotypage	**Caractéristiques**	**Cancers ciblés**
iCOGS	Environ 200 000 SNPs sélectionnés	Cancer du sein et divers types de cancer
OncoArray	Environ 500 000 SNPs sélectionnés	Cancer du sein, de la prostate, du poumon, du colon, de l'ovaire et études génomiques plus larges

https://pgtb.fr/genotypage-de-snp-et-indel-2/

L'imputation génomique est une technique statistique utilisée pour prédire, ou "imputer", les valeurs manquantes dans les domées génomiques. Même avec les technologies de génotypage avancées, il est courant de ne pas obtenir de données pour certains SNPs (polymorphismes mononucléotidiques) chez certains individus. L'imputation peut aider à combler ces lacunes.

Dans cette étude, les chercheurs ont utilisé la version Phase 3 de la base de données 1000 Genomes comme référence pour l'imputation. La base de données 1000 Genomes est un projet international qui a séquencé les génomes de plus de 1000 individus provenant de différentes populations à travers le monde. Elle vise à fournir une représentation détaillée de la variation génétique humaine. En utilisant cette base de données comme référence, les chercheurs peuvent prédire de manière plus précise les valeurs manquantes dans leurs propres données.

SHAPEIT est un logiciel qui "phase" les données génomiques. La phasage est un processus qui détermine quelles variantes spécifiques se trouvent sur le même brin d'ADN. C'est une étape importante avant l'imputation car elle permet d'améliorer l'exactitude des prédictions.

IMPUTE2 est un logiciel qui effectue réellement l'imputation. Il utilise des modèles statistiques pour prédire les valeurs manquantes sur la base des informations disponibles dans les données de référence (dans ce cas, la base de données 1000 Genomes).